

∞ MMLA: HOW MEMORY LETS THE PAST SHAPE THE FUTURE

TECHNICAL REPORT OF MMLA

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ABSTRACT

Proposal. Extending a context window is not the same as learning from a stream. MMLA formalizes a bounded resident memory between transient context and slow weight updates. A completed local segment is eventized; for each event, a neural constructor proposes semantic content and a trusted assembler produces a complete versioned row; deployment either commits that row atomically or returns NULL. Realized futures may price actions during training, while deployment remains causal and future-blind.

Validated components. Controlled-language lifecycle execution is exact on 300/300 held-out records for each of three seeds. Calibrated selection with full-archive fallback improves over a weak budget-matched dense baseline by 5.5–16.6 F1 and over BM25 by 4.0–6.2 F1; the original Llama budget execution is retained as failed, while the corrected Qwen packer satisfies the stated per-record caps. Typed anchor–filler transport is exact on 240/240 controlled held-out records per seed while three same-checkpoint controls obtain no whole-record successes.

Current blocker. The integration loop is incomplete. External, raw-hidden, dense-row, structured-span, and checkpoint-native interfaces fail their preregistered semantic qualifications. In the latest matched comparison, candidate exactness is 0–45/23,040, query exactness is 10–1,536/9,216, and record-macro Brier is 0.998676–0.999093, near the 1,025-class uniform reference 0.999024. Structural mapping passes, but none of the nine jobs qualifies. Predictive overwrite is therefore closed by gate. This report retains the complete evidence and audit record while correcting the formal row, controller, security, and roadmap contracts.

MMLA: architecture, admitted evidence, and the closed loop

four layers are kept separate: formal architecture, implemented ancestors, validated components, and unverified successors

The diagram separates the proposed architecture from admitted component evidence and unresolved integration.

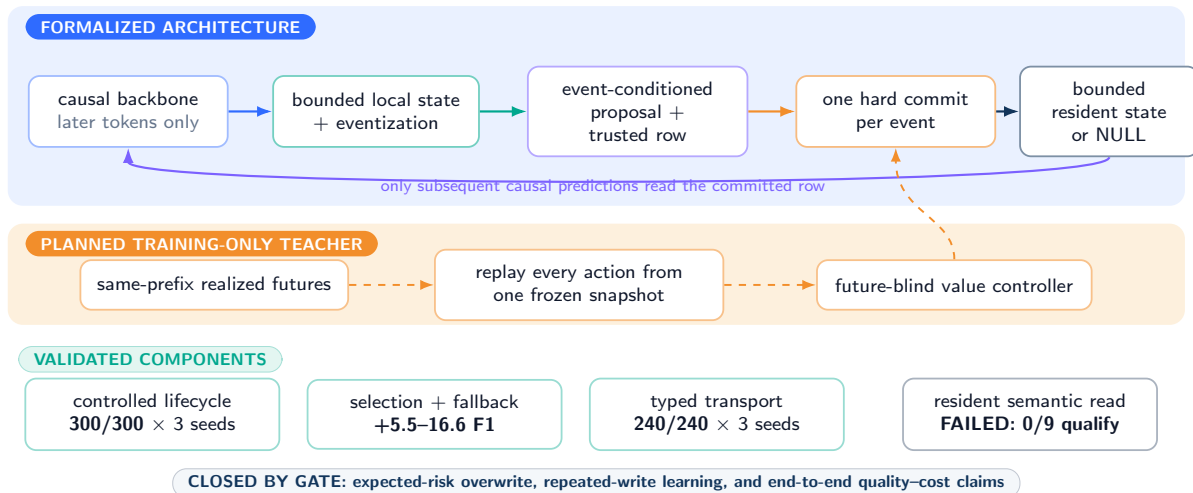


Figure 1: **Proposal, evidence, and present boundary in one view.** Solid arrows show the deployed causal path; the orange branch exists only during training. Admitted component evidence and the matched semantic-interface failure are distinct from the complete predictive-overwrite loop, which has not yet been validated.

Reading guide. This complete companion report uses the same event-level, trusted-assembly, and hard-commit architecture as the submitted arXiv v3 replacement, [arXiv:2606.28876](https://arxiv.org/abs/2606.28876). It preserves the full derivations, experiment history, registered negative results, audits, and appendices. No new learned result is introduced: PM-I2’s nine qualification bundles and formal no-interface terminal remain the latest admitted semantic-interface evidence, and predictive overwrite remains closed. The public project page is <https://github.com/MMLA-org/mmla-memory>.

1 Introduction

We retain the project name MMLA from the earlier *Memory-Managed Long-Context Attention* formulation. The present architecture is mixer-agnostic and is not restricted to attention alone.

Efficient long-context modeling has progressed from the Transformer [56] through linear attention, structured state spaces, recurrent and convolutional hybrids, and sparse attention [29, 20, 54, 45, 19, 46, 61, 64, 63, 16]. These methods reduce the per-token state size or the number of attended positions. They do not, by themselves, answer a different question: when should an observation change what the model will carry forward?

A million-token prompt can still contain mutually inconsistent versions of the same fact, preserve an obsolete value, or force the model to repeatedly process material whose relevance was impossible to know when it arrived. Conversely, a small persistent state can be useful if it supports reliable revision, deletion, abstention, and later reuse. The central object of this report is therefore not an unlimited context window. It is a bounded state that learns to retain the parts of the past that improve future behavior.

The difficult decision is the write, not the read. Immediate salience is a poor proxy for future utility: an ordinary sentence may become important only after later events, while a vivid sentence may never be used again. Training data, however, contains the future that inference lacks. This suggests a division of labor. Real subsequent trajectories can price alternative memory actions; a deployable controller must predict their *conditional expected* cost from completed history alone. Predictive information and information-bottleneck principles supply the target: compress the past while retaining information about the future [55, 7]. The expectation is essential. A policy fitted to the action preferred by one realized continuation need not minimize risk over the future distribution faced at deployment.

We call the resulting construction *predictive memory consolidation*. The model first predicts causally. After a local event, sentence, or chunk has fully arrived, a small bidirectional module may reinterpret that completed segment without changing any output already emitted. It eventizes the segment into a bounded ordered list. For each event, a target-conditioned neural proposal and trusted assembler construct a complete candidate row; a controller chooses one resident target to overwrite or chooses NULL. Insert, update, consolidating update, eviction, deletion, protection, and stale rejection are lifecycle interpretations of this per-event operator. Merging two authoritative resident rows requires two ordered events or a future bounded transaction. The observation horizon and the mixture of local attention, recurrent state, and memory read are deployment choices rather than architectural constants; neither a fixed 1:3:9 ratio nor a fixed 50-turn delay defines the model.

Earlier versions of this report treated lifecycle, fallback, and relational compilation as the endpoint. We now treat them as evidence-bearing components of a broader memory architecture. Controlled lifecycle text exercises write–manage–read–fallback–generate behavior. Held-out multi-hop QA measures the point at which a query-independent writer should abstain. A third study isolates role binding when content and position are unreliable. The new formulation connects these pieces through future-valued consolidation while preserving every earlier result and failure.

Contributions.

1. We specify three bounded state scales—causal token state, completed-segment workspace, and persistent rows—plus slower model parameters. Its only authoritative transition is one complete-row overwrite or NULL per ordered event (§2).
2. We define overwrite selection as conditional action-value estimation. Grouped continuations with identical observed history provide counterfactual branch risks; a causal value predictor sees only completed history and chooses the lowest-risk feasible action. We state the no-leakage interface, information-theoretic objective, stability limits, and system boundary explicitly.
3. We separate representation qualification from predictive overwrite learning. Three frozen-checkpoint semantic-interface studies are registered negatives. In the latest matched comparison, all nine dense-row, structured-span, and checkpoint-native jobs completed and failed the frozen semantic qualification while retaining their structural mapping checks. The conditional expected-risk intervention remains unopened because no interface qualified.
4. Three completed studies provide component evidence: bounded lifecycle execution (§4), calibrated selection and fallback (§5–6), and same-checkpoint typed relational binding (§7).

We do not claim that predictive consolidation is already better than immediate writing, that its information-theoretic form proves intelligence, or that a biological system uses this implementation. The first two studies use frozen language models, and the relational study trains a compact synthetic compiler. A full claim first requires a semantically usable current-state interface and then a corrected same-checkpoint expected-risk intervention; compute-matched pretraining is conditional on both results.

Claim–evidence map: one vocabulary for every layer

status describes the strongest justified statement; it does not transfer upward to the full architecture

CLAIM	ADMITTED BASIS	STATUS
Causal completed-segment boundary	equations + interface contract; no deploy future	FORMALIZED
Controlled lifecycle executor	version, lock, tombstone, eviction implementation	IMPLEMENTED
Lifecycle and typed binding	300/300 and 240/240 per seed under controls	VALIDATED
Resident-row semantic read	PM-S1, PM-I1, PM-I2; PM-I2 has 0/9 qualified	FAILED
Expected-risk overwrite controller	semantic prerequisite failed; oracle never opened	CLOSED BY GATE
Dual-view authoritative row	recommended vNext contract motivated by negatives	PLANNED
Full-system quality–cost claim	no repeated-write or full-system crossover test	PLANNED

The three semantic-interface failures close predictive overwrite without implying that the entire MMLA research program has failed.

Figure 2: **Claim–evidence matrix.** Status belongs to the strongest justified statement, not to the architecture as a whole. “Implemented” denotes a mechanical ancestor; “validated” requires evidence admitted under the frozen protocol.

Claim	Accepted basis	Boundary	Status
Causal boundary	Causal factorization and completed-segment interface	Formal contract only	FORMALIZED
Controlled lifecycle	Version, lock, tombstone, eviction executor	Mechanical ancestor	IMPLEMENTED
Lifecycle and typed binding	300/300 and 240/240 per seed under controls	Controlled grammars	VALIDATED
Resident semantic read	PM-S1, PM-I1, PM-I2; PM-I2 0/9 qualifies	Frozen interfaces/budgets	FAILED
Expected-risk overwrite	Semantic prerequisite failed; oracle unopened	No learned policy result	CLOSED BY GATE
End-to-end cost advantage	No repeated-write or quality–cost crossover	Asymptotic motivation only	PLANNED

Table 1: One status vocabulary separates the formal architecture, implemented ancestors, validated components, failed interfaces, planned work, and work closed by gate.

2 Predictive Memory Consolidation

2.1 Three state scales and a slower parameter scale

Let $x_1, x_2, \dots$ be a token stream. The model maintains a causal hidden state h_t for immediate prediction, a completed local workspace B_n for one consolidation boundary, and exactly S resident row cells. The authoritative resident state is

$$M_n = (r_{n,1}, \dots, r_{n,S}) \in \mathcal{R}_B^S, \quad \mathcal{R}_B = \mathcal{I}_{B_i} \times \mathcal{C}_{L_c} \times \mathcal{V}_d \times \mathcal{K}_{d_k} \times \mathcal{U}_{B_m}, \quad (1)$$

where $\mathcal{I}_{B_i}$ is a fixed-width identity field, $\mathcal{C}_{L_c}$ is a padded canonical encoding over one fixed alphabet Σ with at most L_c symbols, $\mathcal{V}_d$ and $\mathcal{K}_{d_k}$ are fixed-width neural payload and routing-key spaces, and $\mathcal{U}_{B_m}$ is a fixed-width metadata record of at most B_m bits. Concretely,

$$B_c = \lceil \log_2(L_c + 1) \rceil + L_c \lceil \log_2(|\Sigma| + 1) \rceil, \quad B_v = d p_v, \quad B_k = d_k p_k,$$

for fixed payload/key precisions p_v, p_k ; padding and the stored length make the canonical serialization unique. Aliases have a fixed per-row count and length within $B_c + B_m$; provenance and rollback fields contain bounded identifiers or digests, never an append-only history. The resident table is therefore bounded by

$$\text{bits}(M_n) \leq S(B_i + B_c + B_v + B_k + B_m) =: B_{\text{resident}}, \quad (2)$$

independently of stream length. A full provenance document, superseded canonical value, rollback snapshot, retrieval corpus, or audit log belongs to a separately charged external ledger A_n , not to M_n . If that ledger grows, only resident state—not

Formal MMLA architecture: ordered events and one hard commit per event

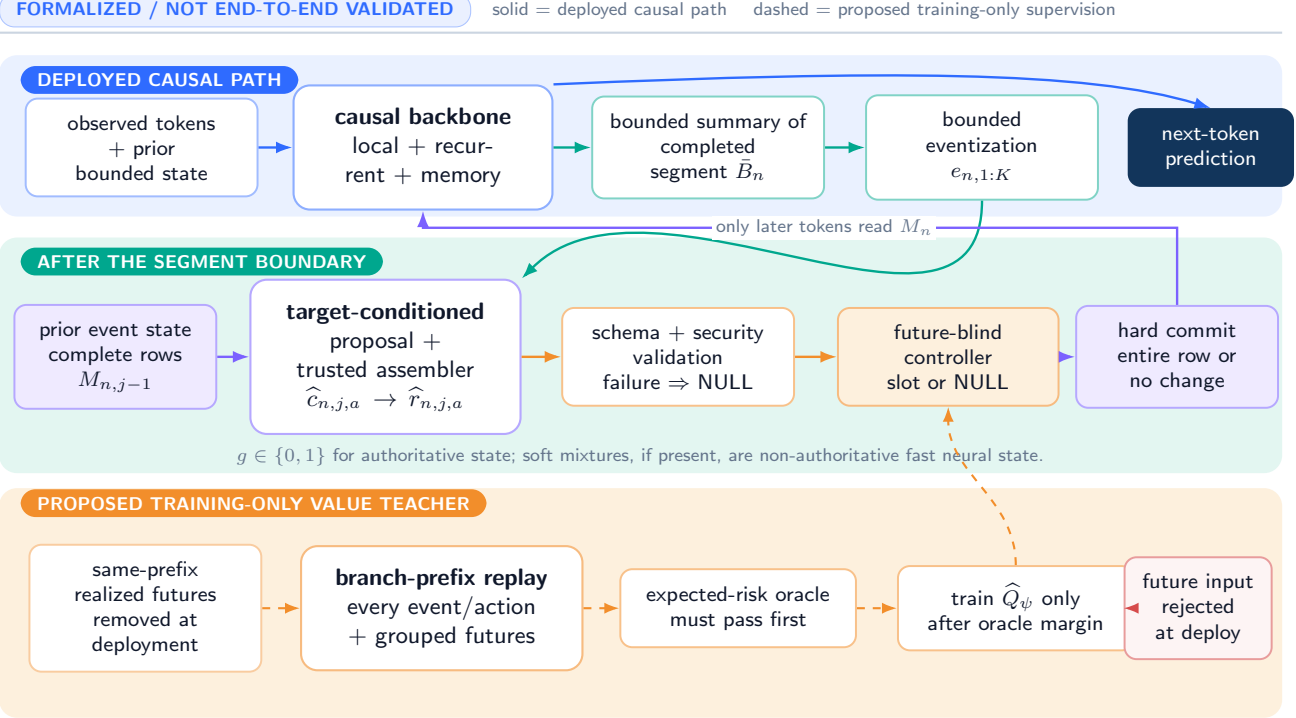


Figure 3: **Predictive memory consolidation.** The language model predicts causally and reads only prior persistent state. After a local segment is complete, a bounded bidirectional consolidator may reinterpret that observed segment and emit ordered events. At training time, several realized continuations with the same prefix estimate the conditional risk of every registered event-level action. A causal value model regresses those risks. At deployment the continuation branches are absent; one complete row is overwritten or NULL leaves state unchanged per event, and only later tokens read the resulting segment state.

the total system—is bounded. Model parameters are a slower fourth time scale: online adaptation changes M_n , whereas conventional training changes the weights.

A block may expose local attention, a recurrent or linear state, and bounded memory read. Their ratio is not fixed. A budget-aware mixture is

$$y_t = \sum_{j \in \{\text{local}, \text{linear}, \text{memory}\}} w_{t,j} y_{t,j}, \quad (3)$$

$$w_t = \text{softmax } r_\theta(h_t, M_{n(t)-1}, b_t),$$

where b_t describes the available compute budget. An implementation may evaluate only the selected branches or place memory blocks sparsely. The invariant is bounded, editable state, not a particular layer ratio.

For a reproducible first end-to-end falsification, we nevertheless fix a proposed *MMLA-R0* reference: 12 mixer blocks in four repetitions of [local attention, recurrent/linear state, memory read], a 2,048-token local window, 256-token completed segments, $S = 256$ resident rows, at most four ordered events per segment, and at most one hard commit per event. The archive is disabled during semantic-interface qualification. Neural computation uses BF16 with FP32 normalization and controller reductions. This is a reference test configuration, not an empirically optimized product or a universal mixer ratio.

2.2 Local bidirectional consolidation without future leakage

Let the n th completed segment occupy $(b_{n-1}, b_n]$. Token prediction remains autoregressive,

$$p_\theta(x_{1:T}) = \prod_{t=1}^T p_\theta(x_t \mid x_{<t}, M_{n(t)-1}). \quad (4)$$

Only after x_{b_n} has been processed does the consolidator form

$$z_n = C_\phi(h_{b_{n-1}+1:b_n}, M_{n-1}), \quad (5)$$

$$M_n \rightarrow p_\theta(x_t \mid x_{<t}) \quad (t \leq b_n).$$

Thus C_ϕ may use bidirectional attention *inside an already observed segment*; its output can affect only later tokens. This is local reinterpretation, not inference-time access to an unobserved future. The segment may be a fixed chunk in the first

experiment and a learned stopping interval later. A maximum horizon is a safety and cost bound, not a claim that consolidation should always wait a fixed number of turns. Equivalently, the deployed consolidator must satisfy

$$z_n \perp X_{>b_n} \mid (X_{\leq b_n}, M_{n-1}), \quad (6)$$

while training-only losses may use $X_{>b_n}$ as a target. This distinction is structural: the forward API contains no future-bearing argument, and the updated state is unavailable to logits at or before the boundary. “Local bidirectionality” therefore means retrospective encoding of observations already received, not bidirectional generation and not a hidden full-sequence look-ahead.

The consolidator next eventizes the bounded segment summary into an ordered list,

$$E_\eta(z_n) = (e_{n,1}, \dots, e_{n,K_n}), \quad K_n \leq K_{\max}. \quad (7)$$

A segment containing several facts is decomposed before one-row actions are considered. Bounded multi-row atomic transactions are deferred; they are not smuggled into a single overwrite.

Within a completed segment, let $M_{n,0} = M_{n-1}$ and process the ordered events sequentially. The state after event j is $M_{n,j}$, and only the completed segment state $M_n = M_{n,K_n}$ becomes available to later tokens.

2.3 One atomic overwrite per event

For event $e_{n,j}$ and target $a \in \{1, \dots, S\}$, a neural constructor proposes semantic content,

$$\hat{c}_{n,j,a} = G_\phi(z_n, r_{n,j-1,a}, M_{n,j-1}, e_{n,j}, a). \quad (8)$$

The proposal contains bounded canonical value and aliases, neural payload, typed routing key, uncertainty, and proposed operation or tombstone intent. A trusted assembler then forms the complete row,

$$\hat{r}_{n,j,a} = \text{AssembleTrusted}(\hat{c}_{n,j,a}, \text{tenant}, \text{ACL}, \text{old protection}, \text{next version}, \text{provenance}, \text{rollback lineage}). \quad (9)$$

Tenant, ACL/security scope, monotone version, protection inheritance, source receipt, rollback lineage, and deletion verification are system-owned fields, not free neural outputs. Their resident representation is fixed-width: receipts and rollback lineage are bounded identifiers into A_n , not embedded histories. Aliases live inside the bounded capsule or in a deterministic derived index; they never form an independently mutable authoritative map. Schema, size, security, and version checks run before action selection. An oversized or invalid candidate fails to NULL. The version counter is an unsigned B_u -bit member of the metadata budget. A write is feasible only when $u_{\text{old}} < 2^{B_u} - 1$ and the trusted assembler sets $u_{\text{new}} = u_{\text{old}} + 1$; exhaustion rejects the write (or requires an explicit external re-key/migration protocol) and never wraps, saturates, or reuses a prior (ι, u) pair.

We recommend a dual-view vNext contract: canonical content makes a row auditable and reproducible, while a synchronized neural payload supports differentiable routing and readout. The two views are not merely co-located. For row identity ι and version u , let c be the canonical view and (v, k) the neural payload and routing key. A deterministic canonicalizer Can and a versioned encoder $E_{\omega,u}$ define the consistency receipt

$$\rho(r) = \left(\iota, u, H(\text{Can}(c)), H(E_{\omega,u}(\text{Can}(c))), H(v \parallel k), H(\omega) \right). \quad (10)$$

A candidate is committable only if its identity/version/security fields agree and the fixed-width serialized neural bytes satisfy

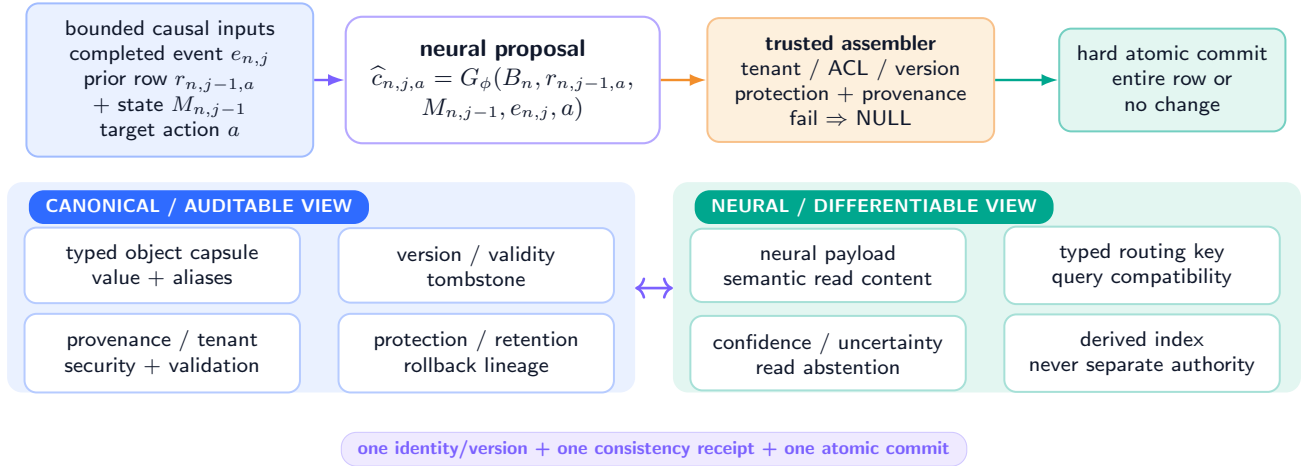
$$\text{Ser}_{v,k}(E_{\omega,u}(\text{Can}(c))) = \text{Ser}_{v,k}(v, k). \quad (11)$$

The hashes in ρ are audit accelerators and tamper-evident receipts, not a substitute for the byte-equality check; a digest collision therefore cannot make unequal resident views valid. Canonical reconstruction and fresh-query semantic gates must also pass under the frozen versioned reader, and the receipt is included in the same atomic commit. The fixed-width $H(\omega)$ identifies an external immutable encoder artifact; encoder weights are not copied into the row. Any later neural re-encoding is a new row version, never an in-place drift. On load, recovery, export, and before use as teacher state, the system rechecks byte equality and ρ ; failure quarantines the row and makes its action infeasible rather than choosing which view to trust. A deterministic derived index is rebuilt only from validated rows and is never authoritative. This closes the formal consistency invariant; whether a learnable encoder can satisfy its semantic gates remains unverified. The three negative semantic-interface studies motivate this contract; they do not validate it.

Recommended vNext interface: a complete dual-view authoritative row

PROPOSED / UNVERIFIED

motivated by three semantic-interface failures; not yet qualified



Commit only if versioned-encoder bytes equal stored neural bytes; hashes are audit receipts and semantic gates also pass. Mismatch quarantines the row; re-encoding creates a new version. Fields are capped; versions never wrap; long provenance stays external.

Figure 4: **Proposed complete-row contract.** Canonical and neural views share one identity, validation boundary, version, and security scope. This vNext interface is explicitly unverified.

For each event j , define the feasible set only after every bounded candidate has been assembled and validated,

$$\mathcal{A}_{n,j} = \{\emptyset\} \cup \{a \in \{1, \dots, S\} : \hat{r}_{n,j,a} \in \mathcal{R}_B, \text{Valid}(\hat{r}_{n,j,a}) = 1\}. \quad (12)$$

The controller chooses $a_{n,j} \in \mathcal{A}_{n,j}$ and a hard authoritative gate $g_{n,j} \in \{0, 1\}$, with $g_{n,j} = 0$ exactly when $a_{n,j} = \emptyset$. The state transition is

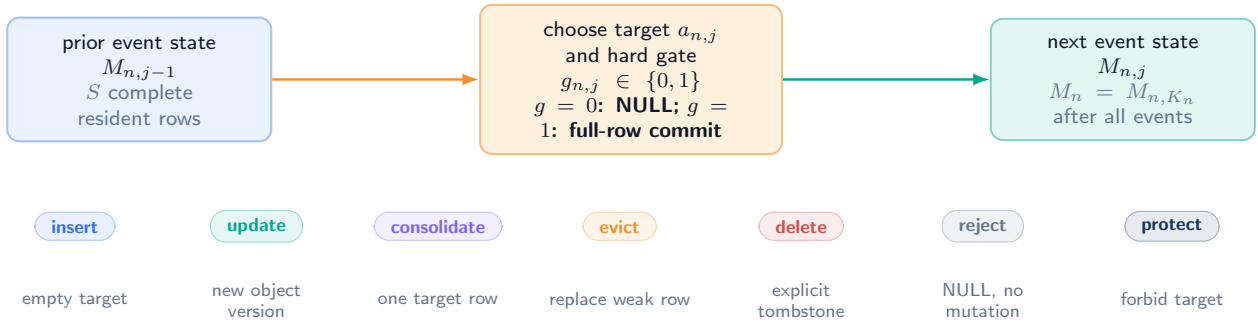
$$M_{n,j} = \begin{cases} M_{n,j-1}, & g_{n,j} = 0 \text{ (NULL)}, \\ \text{Replace}(M_{n,j-1}, a_{n,j}, \hat{r}_{n,j,a_{n,j}}), & g_{n,j} = 1, \end{cases} \quad M_n = M_{n,K_n}. \quad (13)$$

Insert, update, consolidating update, eviction, and deletion are lifecycle interpretations of replacing one complete row. Merging two authoritative resident rows requires two ordered events or a future bounded transaction, because changing one target cannot silently invalidate another row. A tombstone is itself a committed, versioned row; NULL changes nothing. Protection removes forbidden actions. A soft mixture, if useful, belongs only to non-authoritative fast neural state and never to the versioned fact table.

Hard event-level transition: one complete row or NULL

the versioned fact table never uses a soft mixture; soft state, if used, is explicitly non-authoritative

canonical capsule + neural payload + typed key + version/validity + protection/retention + provenance/security + tombstone commit together



Aliases live inside the object capsule or a deterministic derived index; no independently mutable authoritative alias map.

Figure 5: **One physical transition, several lifecycle meanings.** The selected target, complete replacement row, and NULL option define the authoritative operation.

If neural payloads satisfy $\|v(r)\|_2 \leq B$, a committed payload change obeys

$$\|v(\widehat{r}_{n,j,a_{n,j}}) - v(r_{n,j-1,a_{n,j}})\|_2 \leq 2B. \quad (14)$$

This is a one-step payload bound, not a convergence theorem. A poor policy can still oscillate or erase useful state. Permanent recovery of every historical version is impossible under fixed resident capacity unless versions are summarized or moved to a separately charged archive.

Worked state transition: Alice moves from Paris to Berlin

ILLUSTRATIVE / NOT MEASURED

numbers show the contract, not experimental evidence

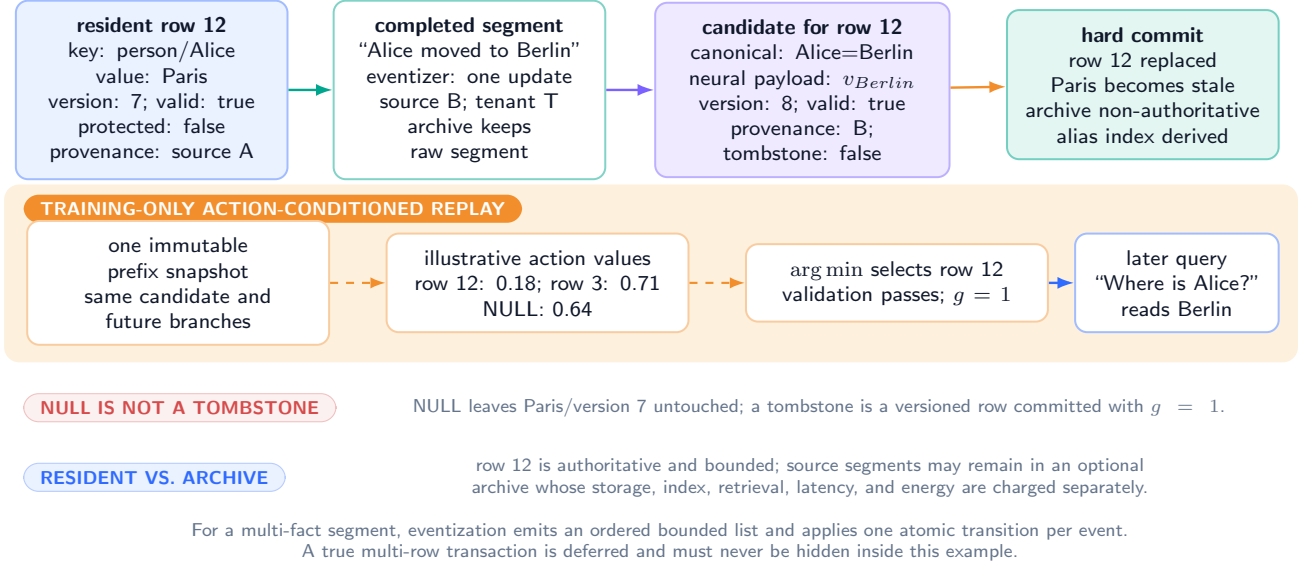


Figure 6: **Worked state transition.** Alice’s authoritative row advances from Paris/version 7 to Berlin/version 8. Numerical action values are illustrative, not measured. NULL preserves version 7; a tombstone would be a committed version 8 row. The replay teacher replays each target-conditioned action from the same snapshot; the archive remains non-authoritative and separately charged.

System and recovery contract. The action mask enforces protection and tenant scope before values are compared. A failed schema, signature, provenance, monotone-version, or security check returns NULL. A committed row and its audit record publish atomically. Recovery restores a prior versioned snapshot and invalidates deterministic derived indexes. Verified deletion writes a tombstone and applies the declared archive-retention policy; it is never conflated with NULL. These are formal requirements, not end-to-end validated mechanisms.

2.4 Multi-event, multi-future risk teacher and causal action values

Let the bounded deployment summary for event j be $s_{n,j} = S_\theta(h_{b_n-w+1:b_n}, z_n, M_{n,j-1}, e_{n,j})$. The notation $x_{\leq b_n}$ describes only the causal information set: deployment cannot reread the full prefix unless archive retrieval and its cost are explicit. A segment is one finite-horizon decision episode with events $j = 1, \dots, K_n$. For any event-action sequence $\mathbf{a}_{1:j} = (a_1, \dots, a_j)$, define the prefix state recursively by

$$M_{n,0}^{()} = M_{n-1}, \quad M_{n,j}^{(\mathbf{a}_{1:j})} = \text{Commit}\left(M_{n,j-1}^{(\mathbf{a}_{1:j-1})}, a_j, \widehat{r}_{n,j,a_j}^{(\mathbf{a}_{1:j-1})}\right). \quad (15)$$

The event summary, candidate, feasible mask, and later event state are recomputed from that branch’s validated prefix; mixing a candidate from one prefix with another prefix is invalid. The future continuation F remains random under $P(F \mid s_{n,1})$.

The exact sequence teacher prices a policy or complete action sequence $\mathbf{a} = (a_1, \dots, a_{K_n})$ from one immutable pre-segment snapshot. A deterministic sequence is a special future-blind policy and is useful for exhaustive enumeration:

$$J_H^{\text{seq}}(\mathbf{a}; s_{n,1}, F) = \sum_{j=1}^{K_n} \lambda_{\text{write}} \mathbf{1}[a_j \neq \emptyset] + \lambda_{\text{int}} D_{\text{collateral}}(M_{n,K_n}^{(\mathbf{a})}) + \sum_{k=1}^H \gamma^{k-1} \ell(F_k; M_{n,K_n}^{(\mathbf{a})}). \quad (16)$$

For a feasible action a at event j after branch prefix $\mathbf{a}_{<j}$, let $\Pi_{j+1:K_n}^{\text{fb}}$ be the preregistered class of deterministic or stochastic continuation policies whose action at event q is measurable only with respect to the completed-segment event history and branch state $s_{n,q}$, never the identity or tokens of sampled F . Its teacher value is the future-blind cost-to-go

$$\mathcal{Q}_{n,j}^T(a \mid s_{n,j}^{(\mathbf{a}_{<j})}) = \min_{\pi \in \Pi_{j+1:K_n}^{\text{fb}}} \mathbb{E}_{F, \mathbf{a}_{j+1:K_n} \sim \pi} [J_H^{\text{seq}}((\mathbf{a}_{<j}, a, \mathbf{a}_{j+1:K_n}); s_{n,1}, F)] . \quad (17)$$

The order of minimization and expectation is essential: moving the minimum inside $\mathbb{E}_F$ would allow a clairvoyant continuation action chosen after seeing the sampled future. Exact dynamic programming minimizes the grouped expected return at every branch prefix; an approximation uses one frozen future-blind rollout policy declared before held-out futures are opened. Thus later events are either explicitly optimized or explicitly governed; they are never silently ignored. When $K_n = 1$, this reduces to the single-event risk below. For the first matched causal intervention, the registered protocol deliberately sets $K_n = 1$ and forbids later writes inside the scoring horizon, so that every branch isolates exactly one action:

$$J_H(a; s_{n,j}, F) = \sum_{k=1}^H \gamma^{k-1} \ell(F_k; M_{n,j}^{(a)}) + \lambda_{\text{int}} D_{\text{collateral}}(a) + \lambda_{\text{write}} \mathbf{1}[a \neq \emptyset], \quad (18)$$

where $M_{n,j}^{(a)}$ is the event-level state after action a . The frozen reader must replay the continuation under that post-action state; adding memory only after a future hidden trace has already been cached is a fixed-trace diagnostic, not an action-conditioned counterfactual. The first matched experiment forbids later writes inside this horizon, so that J_H identifies the consequence of exactly one intervention. The collateral term measures damage to facts that were not modified, leakage of obsolete values, or destruction of a still-useful slot.

The deployable single-event objective is the conditional action value

$$\bar{J}_H(a \mid s_{n,j}) = \mathbb{E}_{F \sim P(\cdot \mid s_{n,j})} [J_H(a; s_{n,j}, F)] . \quad (19)$$

For either the sequence return in Eq. 16 or its registered $K_n = 1$ restriction, the formal benchmark approximates the future expectation with K_F physically grouped continuations sharing the identical prefix and pre-decision memory snapshot,

$$\tilde{J}_H(a \mid s_{n,j}) = \sum_{k=1}^{K_F} w_k J_H(a; s_{n,j}, F_k), \quad \sum_k w_k = 1, \quad (20)$$

with fixed preregistered weights w_k that are normalized within a group and never inferred from the preferred action. Group IDs, not continuation branches, are split across fit, calibration, and held-out data. For $K_n > 1$, the same grouped futures price every branch prefix in the exact dynamic program or the declared frozen rollout policy. A future-blind value model $\hat{Q}_\psi(a \mid s_{n,j})$ regresses the detached cost-to-go target for *every* feasible action, for example with a Huber loss,

$$\mathcal{L}_Q = \sum_{a \in \mathcal{A}_{n,j}} \text{Huber}(\hat{Q}_\psi(a \mid s_{n,j}) - \text{stopgrad } \tilde{Q}_{n,j}^T(a \mid s_{n,j})) . \quad (21)$$

At deployment it processes events in order, choosing $\arg \min_a \hat{Q}_\psi(a \mid s_{n,j})$, committing the chosen row, recomputing $s_{n,j+1}$ from the resulting state, and optionally requiring a calibrated margin over NULL. It receives no continuation, future hidden state, answer, or gold action. In the first causal intervention, $K_n = 1$ and one payload with $g_{n,1} = 1$ may deliberately be shared across slot actions to isolate *where or whether to write*. That controlled simplification is not the complete target-conditioned architecture of Equations 8–9 and cannot establish multi-event credit assignment.

This distinction repairs a failure of the earlier formulation. In general,

$$\mathbb{E}_F [\text{softmax}(-J_F/\tau)] \neq \text{softmax}(-\mathbb{E}_F [J_F]/\tau) . \quad (22)$$

Consequently, KL imitation of a Gibbs policy built from one realized future does not by itself optimize conditional expected risk. The earlier code remains useful for interface qualification, but it is not the decisive predictive-consolidation estimator. A learned latent rollout may later approximate $P(F \mid s_{n,j})$, as in world-model and latent-imagination systems [21, 22]; it is not the first source of supervision, because a misspecified predictor can reward memories that merely make its own imagined future self-consistent.

2.5 Motivation only: predictive information under finite capacity

The persistent state is a lossy code of the observed past. We want it to approach a sufficient statistic for the future,

$$P(X_{>b_n} \mid X_{\leq b_n}) \approx P(X_{>b_n} \mid M_n), \quad (23)$$

rather than reconstruct every previous token. A predictive information-bottleneck objective is

$$\max_{q(M|X_{\leq b_n})} I(M; X_{>b_n}) - \beta I(M; X_{\leq b_n}) - \lambda \mathbb{E}[N_{\text{overwrite}}]. \quad (24)$$

This is the information-bottleneck principle with future observations as the relevant variable [55, 7]. If the deployed slots use a finite representation of b bits each, then

$$\begin{aligned} I(M; X_{\leq b_n}) &\leq H(M) \leq Sb, \\ I(M; X_{>b_n}) &\leq I(X_{\leq b_n}; X_{>b_n}), \end{aligned} \quad (25)$$

where the second inequality follows from data processing. The architecture cannot create predictive information or evade finite storage; it can only learn which predictive information to preserve. A variational implementation may replace the mutual-information penalty with a KL term to a bounded prior [2].

For the finite action set, let $E(a) = \bar{J}_H(a | s_{n,j})$ and $Z = \sum_a \exp[-E(a)/\tau]$. Any entropy-regularized action distribution q obeys the exact identity

$$\begin{aligned} \mathcal{F}[q] &= \mathbb{E}_q[E(a)] - \tau H(q) \\ &= \tau D_{\text{KL}}(q \| q^*) - \tau \log Z, \\ q^*(a) &= \frac{e^{-E(a)/\tau}}{Z}. \end{aligned} \quad (26)$$

Equation 26 is a regularized decision identity for one *already averaged* conditional-risk vector. It does not justify averaging Gibbs policies formed separately for sampled futures; Eq. 22 shows why. The first intervention trains action values directly. Here τ is an optional optimization temperature, not the physical temperature of a GPU or a brain.

For completeness, an overwrite is logically irreversible when it discards the previous slot. Landauer’s ideal lower bound is

$$Q_{\min} \geq b_{\text{erase}} k_B T \ln 2 \quad (27)$$

for erasing b_{erase} independent bits [33]. This observation has no evidential role in the architecture or experiments. It establishes compatibility with thermodynamic accounting, not measured energy efficiency; present hardware operates far above this limit, and fewer logical state changes need not reduce wall-plug power.

2.6 Bounded resident state and the full system cost

Let T be total stream length, W the local window, S the resident slots, $K_r \ll S$ the active slots read per token, K_e the maximum events per segment, and C the minimum tokens per consolidation. Let $R_q(S, d)$ be the routing cost for one token, $C_{\text{cons}}(W, d)$ the completed-segment encoding/eventization cost, $C_G(d, L_c)$ the cost of constructing one target-conditioned semantic proposal, $C_A(B_m)$ the trusted assembly/dual-view validation cost per candidate, $C_Q(S, d)$ the action-scoring cost per event after candidates exist, and $C_{\text{commit}}(B_{\text{row}})$ the bounded row-commit cost. If the complete architecture materializes all S target candidates before choosing an action, the honest deployment context-mixing budget is

$$\begin{aligned} C_{\text{deploy}}(T) &= O(TWd) + O(TR_q(S, d)) + O(TK_r d) \\ &\quad + O\left(\frac{T}{C} [C_{\text{cons}}(W, d) + K_e (S(C_G + C_A) + C_Q + C_{\text{commit}})]\right), \\ \text{bits}(M_n) &\leq B_{\text{resident}} = S(B_i + B_c + B_v + B_k + B_m). \end{aligned} \quad (28)$$

Here C_G and C_A are arguments of their displayed shapes; the shorthand suppresses them only inside the bracket. A shared precomputation may reduce SC_G to $C_G^{\text{shared}} + SC_G^{\text{target}}$, and a preregistered target shortlist of size $L \ll S$ may replace S only if shortlist construction, misses, and excluded actions are charged and reported. Neither optimization may omit complete-candidate construction from accounting. An exact read scan has $R_q(S, d) = \Theta(Sd)$. An approximate index can reduce query work only by adding explicit build, update, storage, and miss costs; those costs do not disappear from the system boundary.

Training cost is larger because the teacher branches actions and futures. For exact multi-event dynamic programming, let $N_{\text{node}} \leq \sum_{j=0}^{K_e-1} (S+1)^j$ be the number of reachable branch prefixes before the next action after feasibility pruning. With K_F grouped futures and replay cost $C_{\text{replay}}(H)$ per complete sequence leaf, let $N_{\text{leaf}} \leq (S+1)^{K_e}$ be the corresponding number of complete feasible action sequences. The additional teacher budget is

$$C_{\text{teacher}} = O\left(\frac{T}{C} [N_{\text{node}} (S(C_G + C_A) + C_Q + N_{\text{child}} C_{\text{commit}}) + N_{\text{leaf}} K_F C_{\text{replay}}(H)]\right). \quad (29)$$

where $N_{\text{child}} \leq S$ is the maximum number of feasible non-NULL children materialized from a prefix (NULL reuses the parent state). The registered first intervention has $K_e = 1$, so $N_{\text{node}} = 1$, $N_{\text{leaf}} = S+1$, and $N_{\text{child}} = S$ before feasibility pruning: all S

complete candidates are built, all S non-NULL successor rows are materialized, and all $S + 1$ actions, including NULL, are replayed across every grouped future. If an implementation uses immutable structural sharing, it may replace SC_{commit} by a separately measured branch-state materialization cost, but it may not charge only the ultimately selected deployment commit. Any multi-event successor must report pruning or rollout-policy approximation separately rather than hiding the exponential branch factor.

Sparse fallback over an archive A_T makes the *resident* state bounded but not the total retained information: archive and audit-ledger size, index size, retrieval latency, and retrieval energy must be reported separately. With fixed W, S, K_e, C and a specified router/candidate policy, online deployment cost remains linear in the stream, but this is not a claim of constant total memory or small constant factors. The bound also remains aspirational until consolidation, candidate construction, validation, routing, and overwrite are fused into a few large kernels; an implementation dominated by small launches can lose despite favorable asymptotics. BF16 is the reference training path, FP8 is appropriate for qualified matrix multiplications, and normalization, accumulated risk, and discrete action values retain the FP32 islands required for stability. The design introduces no FP64 requirement. FP4 is a later inference or post-training option whose overwrite stability must be tested separately.

2.7 The registered minimal intervention

The next comparison freezes one completed small-model checkpoint and its reader, then trains only parameter-matched consolidation/controller modules on new, physically separated data. Four deployable arms and one oracle share the state size, candidate payload, gate, action candidates, overwrite budget, future-blind deployment interface, optimization exposure, and final checkpoint:

Arm	Segment representation	Write supervision
No-op memory	causal	always NULL
Immediate writer	causal final state	current-only risk
Local consolidation	bidirectional completed segment	current-only risk
Expected-risk consolidation	same local consolidator	grouped multi-future action-value regression
Expected-risk oracle	same local consolidator	empirical conditional risk at evaluation; non-deployable ceiling

The immediate-to-local difference isolates post-segment reinterpretation. The local-to-expected-risk difference isolates the use of a continuation distribution rather than current-only risk. Every benchmark group contains an identical observed prefix with four to eight continuations, including ambiguous prefixes for which NULL is optimal in expectation and cases whose value appears only later. No later write is allowed during the scoring horizon. The oracle first tests whether the data and action set contain useful conditional future value; if it does not beat current-risk writing, scale cannot rescue this intervention. Student evaluation reports future loss, action regret, strict overwrite success, stale-value leakage, preservation of untouched facts, overwrite rate, latency, resident and archive memory, routing/index cost, kernel count, and energy.

2.8 Evidence-gated training allocation

The proposed memory capability is primarily a *mid-training or post-training* intervention, not a reason to begin with an expensive from-scratch model. Let θ_0 be one frozen backbone, ω a current-state interface, and ψ the future-blind action-value model. The evidence order is

$$\begin{aligned}\hat{\omega} &= \arg \min_{\omega} \mathcal{L}_{\text{interface}}(\omega; \theta_0), & \theta &= \theta_0, \\ \hat{\psi} &= \arg \min_{\psi} \mathcal{L}_Q(\psi; \theta_0, \hat{\omega}), & \theta &= \theta_0.\end{aligned}\tag{30}$$

The first line asks whether the resident state can be read and reconstructed at all; the second asks whether a deployable controller can exploit that interface to reduce future risk. Only if both fixed-checkpoint stages beat matched controls without collateral damage does joint training become scientifically justified:

$$\min_{\theta, \omega, \psi} \mathcal{L}_{\text{LM}}(\theta) + \lambda_{\text{mem}} \mathcal{L}_{\text{interface}} + \lambda_Q \mathcal{L}_Q.\tag{31}$$

Equation 31 is a conditional scale study, not a completed result. This sequence keeps the comparatively cheap representation and policy hypotheses falsifiable before allocating a full pretraining budget.

Current evidence boundary. An earlier one-realization CPU loop remains a useful interface demonstration, but its cached future and realization-specific Gibbs target do not estimate Eq. 19. Three subsequent three-seed current-state qualifications failed their preregistered semantic conditions. The first used an external semantic adapter; the second asked whether a raw hidden representation was directly usable. The third compared a dense row, a structured 4×192 span, and the checkpoint-native reader under matched data, training, and final-only evaluation. A versioned repair corrected the mismatch between conceptual audit names and physical safetensors keys before any successful training result existed. All nine repaired jobs then completed

The query arrives only after bounded state construction

write above the boundary · read, abstain, and generate below it

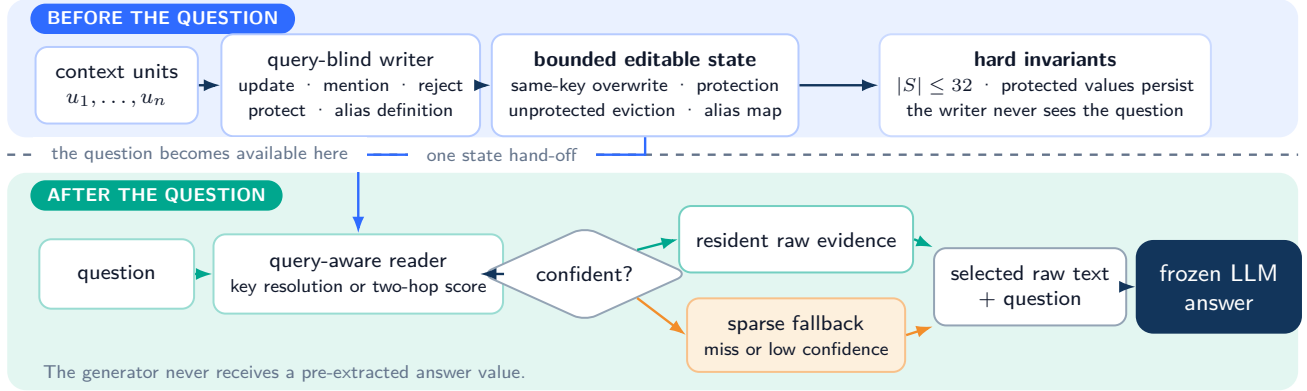


Figure 7: The implemented path. State construction finishes before the question crosses the dashed boundary. At query time, the reader either uses resident evidence or abstains to budgeted sparse retrieval. The frozen language model receives the selected raw text and the question, never a pre-extracted answer value.

2,048 updates, froze predictions before evaluator truth, and were evaluated unconditionally. Candidate exactness remained 0–45/23,040, query exactness 10–1,536/9,216, and record-macro Brier 0.998676–0.999093. Old-value suppression and physical-map invariance passed, but every semantic qualification failed. This localizes rather than erases the problem: the registered readers preserve row identity yet do not expose usable current content. No arm comparison is admissible, and expected-risk overwrite training remains unopened.

The analogy to complementary learning systems is useful but limited: fast editable state and slow parameter learning play different computational roles [40]. We do not claim fixed slots, atomic overwrite, or conditional-risk supervision as a biological model. More generally, a system that can maintain revisable world state under a fixed budget could be an important substrate for persistent agents. That conditional possibility is narrower than a claim of artificial general intelligence: planning, goals, exploration, causal modeling, social reasoning, safety, and slow continual learning remain separate problems.

3 Implemented Evidence Path

3.1 Components

Let a request context be segmented into candidate units $u_1, \dots, u_n$ (event lines in the lifecycle study; whole passages in the multi-hop QA study). The path executes in the following order.

Writer (query-independent). A classifier scores each unit from unit-local features only; its interface does not accept the final question. A unit test enforces this restriction, and identifiers, values, and digits are masked in the controlled lifecycle features. In that study the writer labels each line as `update`, `mention`, `reject`, `lock`, or `alias-def`. In the multi-hop QA study it predicts passage salience.

Controlled-language lifecycle executor. Units stream in document order into a memory of at most 32 slots, keyed by entity in the lifecycle study and ranked by salience in the QA study. `update` writes; `lock` writes and protects, making later writes to that key void; `reject` and `mention` never write; alias definitions populate a separate alias map. When a new key arrives at capacity, the least-recently-updated unprotected slot is evicted. This executor tests lifecycle semantics under controlled cues. Because its alias map is independently mutable, it is not itself an implementation of the single-row atomic state contract in Eq. 13.

Reader (query-aware). At question time the reader either resolves the requested key, directly or through the alias map, or scores resident passages with a learned two-hop reranker. The reranker combines dense-retrieval-style features [28], BM25 [49], lexical overlap, and bridge features. Its dense feature is a frozen-language-model hidden state, not a separately trained DPR encoder.

Calibrated sparse fallback. A confidence signal — the writer probability of a slot’s source line in the lifecycle study, or the mean of the two highest resident-passages scores in the QA study — is compared with a threshold chosen from calibration data only. A miss or a score below that threshold sends the request to budget-matched sparse retrieval over the full context. The threshold must be calibrated at the intended context length. When it was calibrated on short contexts, where 32 slots contain every passage, fallback never fired and extended-length evidence coverage fell from 0.88 to 0.31 (§10).

Generation. A frozen instruction-tuned model receives only the *raw selected evidence* (verbatim context lines/passages — asserted at prompt-build time, never a pre-extracted answer value) plus the question, and generates the answer greedily.

The state transition and the fallback decision can be written compactly. Let $S_t = \{(k_i, v_i, p_i, \rho_i)\}_{i=1}^{m_t}$ contain each resident

key, value, protection bit, and retention score. For a writer action a_t , the lifecycle operator obeys

$$\begin{aligned} S_{t+1} &= \mathcal{L}(S_t, u_t, a_t), \\ |S_{t+1}| &\leq M, \\ p_i = 1 &\Rightarrow (k_i, v_i) \text{ is protected.} \end{aligned} \tag{32}$$

After the question q becomes available, the evidence passed to generation is

$$E(q, S_n, x) = \begin{cases} R(q, S_n), & \text{hit, } c(q, S_n) \geq \tau, \\ \text{SparseRetrieve}(q, x; B), & \text{otherwise,} \end{cases} \tag{33}$$

where B is the fixed evidence budget and τ is chosen on calibration data. Equations 32–33 expose the two guarantees that are easy to conflate: the lifecycle constrains how resident state may change, while fallback handles evidence that was absent from that state.

Two instantiations, disclosed. The controlled versioning study exercises typed events, same-key overwrite, protection with void writes, LRU eviction, and alias resolution. The multi-hop QA study contains static encyclopedia passages, so its memory reduces to a *bounded salience cache*: the 32 passages with the highest query-independent writer scores. No overwrite or protection event fires in that setting. Its writer is query-free at inference, although its training labels come from question-linked supporting facts in the training split. These differences limit what the real-text result can say about lifecycle semantics.

3.2 How the completed evidence maps to the proposal

The implemented path predates the conditional-risk teacher and does not instantiate the entire model in Figure 3. Its role is to test semantics that a learned overwrite must eventually preserve. The controlled versioning study exercises same-key update, protection, stale rejection, eviction, and NULL-like non-writes under a controlled grammar. The multi-hop QA study measures the writer’s information boundary and the value of abstention. The relational study tests a candidate-construction side path before lifecycle execution. None compares immediate writing with local or expected-risk consolidation.

A separate native scaffold uses pre-norm RMSNorm, rotary embeddings, grouped-query local attention, SwiGLU, and intermittent memory-managed mixers [65, 53, 1, 26, 52]. It has completed a 50,003,968-token real-data pilot. Predictive consolidation and conditional-risk supervision were not part of that training, so the run is evidence of trainability and checkpoint availability rather than evidence for the new mechanism. The staged studies in Eq. 30 reuse its final checkpoint precisely to avoid attributing backbone or data-order differences to memory-interface learning.

The first CPU implementation enforces useful interface invariants: reads depend only on prior persistent state; local consolidation sees only a completed segment; deployment methods reject future-bearing arguments; and a hard update changes no more than one payload row. It does not replay future tokens under each action, estimate risk from multiple continuations with the same prefix, or include all lifecycle metadata in the resident row. It is therefore a mechanical ancestor, not positive evidence for the learned architecture.

Interface	Seed	Candidate exact	Query exact	Mean Brier	Preservation	Physical map
External semantic adapter	2026082101	358/6656	244/2048	0.6490	13/1536	pass
	2026082102	314/6656	441/2048	0.6263	3/1536	pass
	2026082103	406/6656	258/2048	0.6342	20/1536	pass
Raw representation probe	2026083101	12/2048	343/131072	1.4201	—	pass
	2026083102	43/2048	263/131072	1.4230	—	pass
	2026083103	36/2048	513/131072	1.4023	—	pass

Table 2: Registered frozen-checkpoint interface negatives. “Physical map” denotes the preregistered invariance checks, not semantic success. The external adapter also passed old-value suppression but failed reconstruction, query, calibration, and unrelated-fact preservation. The raw probe failed its first five semantic conditions and passed only two mapping conditions on every seed. These experiments localize an interface problem; they do not reject the native memory mechanism or predictive consolidation.

Method	Held-out exact match (3 seeds)	Purpose
Lexical first-key	0.087	floor control
Lexical last-key	0.333	strongest non-learned baseline
Naive all-writes lifecycle	0.333	no event typing
Writer allowed to see the question	0.333	leakage control
Learned writer; lifecycle disabled	0.490/0.490/0.493	component removal
Lifecycle active; fallback disabled	0.990 (all seeds)	component removal
Complete controlled lifecycle path	1.000 (all seeds)	proposed path
Lifecycle with oracle event labels	1.000	diagnostic ceiling

Table 4: Controlled-language lifecycle results on 300 held-out records, read once. Before the split was opened, we required improvement over the strongest hand-written rule on every seed with a positive paired-bootstrap 95% confidence interval, substantially fewer lifecycle errors than the learned writer without lifecycle control, at least 0.90 accuracy on protected values, recovery of most capacity misses, and no more than 32 resident slots. The complete path met each condition: a +66.7-point margin with 95% CI [+61.3, +72.0], zero lifecycle errors versus 0.51, protected-value accuracy 1.0, recovery of all 3/3 target evictions, and peak occupancy 32. These numbers validate controlled execution, not learned natural-language memory management.

Seed	Reader	Candidate exact	Query exact	Missing → absent	Mean Brier	Map	Qualified
2026091101	D1 dense row	0/23040	1536/9216	1536/1536	0.998981	pass	no
	S1 structured span	15/23040	1536/9216	1536/1536	0.998676	pass	no
	N1 native reader	30/23040	645/9216	640/1536	0.998992	pass	no
2026091102	S1 structured span	21/23040	10/9216	0/1536	0.999035	pass	no
	N1 native reader	45/23040	10/9216	0/1536	0.999046	pass	no
	D1 dense row	0/23040	1536/9216	1536/1536	0.998815	pass	no
2026091103	N1 native reader	30/23040	10/9216	0/1536	0.999093	pass	no
	D1 dense row	0/23040	1536/9216	1536/1536	0.998818	pass	no
	S1 structured span	0/23040	1536/9216	1536/1536	0.998977	pass	no

Table 3: PM-I2 current-interface qualification from one frozen V28 release. Predictions were materialized before evaluator truth; all 55,296 record rows were separately reduced. Mean Brier is near the 1,025-class uniform reference $1024/1025 = 0.999024$. Every job passes old-value suppression and the registered 64-row mapping checks, but none passes the semantic conditions. Because no interface qualifies at all three seeds, no arm comparison or preferred-reader claim is defined. A separately implemented within-project adjudicator publishes the registered no-interface terminal and marks bootstrap comparison not applicable.

PM-I2 moves one level closer to the model than Table 2: D1 reads a dense row, S1 preserves a structured 4×192 span, and N1 calls the checkpoint-native memory path. Table 3 shows that the result is not a narrow failure of one external codec. The nine jobs preserve the registered structural invariants yet remain close to chance on semantic output. The comparison therefore rules out these three frozen interfaces under their matched budgets; it does not show that the V28 backbone lacks potentially useful information, nor does it distinguish an optimization bottleneck from an unsuitable compositional output geometry. A first launch-only package stopped before production because a post-freeze test used the receipt builder at the wrong lifecycle stage. A versioned repair changed only that lifecycle test and launch identity; the separately implemented within-project adjudicator then reopened the admitted factors and truth, reproduced all nine false decisions, published the registered no-interface-qualified terminal, and performed no bootstrap draws. The exact machine identifier is retained in Appendix H and the evidence-admission audit.

3.3 Preregistration and test discipline

Before generating confirmatory data, we wrote down the comparisons, the numerical conditions that would count as support, and the circumstances that would stop the run. Frozen holdouts were then read once; the multi-hop QA holdouts were read once for each of the two reader models specified in advance. Completion markers prevent accidental re-execution, and every artifact records the code revision that produced it. An earlier version of the lifecycle task was discarded before its test split was read because a lexical rule already solved every example. Section 8 preserves this history and the later evaluator corrections.

4 Controlled Lifecycle Study

Data. 1,200/300/300 train/dev/test records (split-disjoint sentence frames, shared cue vocabulary — a disclosed controlled-language boundary), six balanced scenarios: *base-overwrite*, *alias-query* (the query names the key only through an alias defined once in context), *stale-remention* (post-final-update audit lines repeat old values), *invalid-write* (rejected change requests propose wrong values), *protected-slot* (a lock freezes the value; later updates are void), and *slot-pressure* (48 distinct written keys exceed the 32-slot bound). Generator tests prove that “take the last line mentioning the queried key” is wrong by construction on the four middle scenarios — and returns exactly the designed wrong value — while remaining correct on the other two.

Table 4 and Figures 8–9 make the component costs visible. Removing event typing or allowing the writer to depend on the question returns accuracy to the 0.333 lexical level. Keeping the learned writer but skipping lifecycle execution yields about

Removing lifecycle control or fallback has a visible cost

exact match · 300 held-out records · mean of three fixed seeds

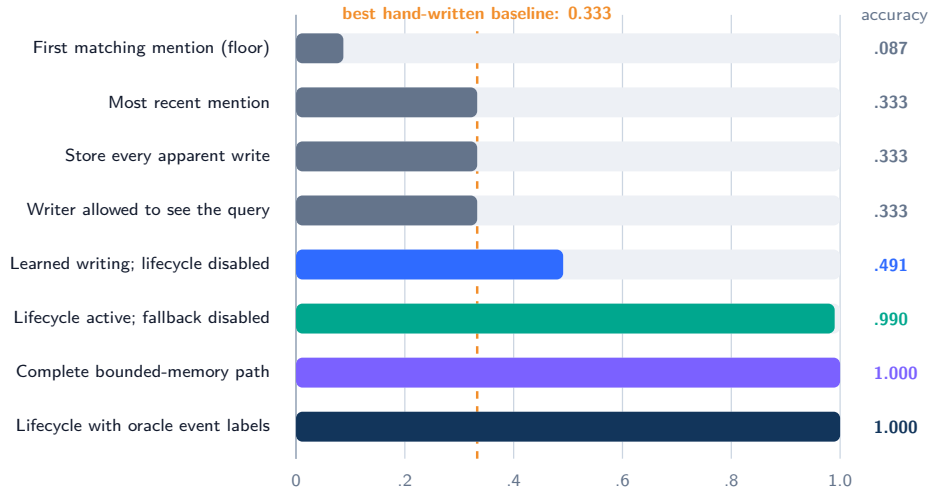


Figure 8: Held-out exact match in the controlled lifecycle study, averaged over three fixed seeds. The dashed line is the strongest hand-written rule. Event typing and query-independent writing are necessary to rise above that rule; disabling lifecycle execution roughly halves accuracy, while disabling fallback loses only the capacity-pressure cases.

Lifecycle failures are scenario-specific by construction

exact match · mean of three seeds

	base overwrite	alias query	stale re-mention	invalid write	protected slot	slot pressure
Lexical last-key	1.00	0.00	0.00	0.00	0.00	1.00
Naive lifecycle	1.00	0.00	0.00	0.00	0.00	1.00
Lifecycle disabled	.37	.44	.39	.41	1.00	.33
Fallback disabled	1.00	1.00	1.00	1.00	1.00	.94
Complete path	1.00	1.00	1.00	1.00	1.00	1.00

retrieval baselines succeed only where last mention is valid full path: correct in all six regimes

Figure 9: Held-out exact match by lifecycle scenario, averaged over three fixed seeds. Lexical and untyped rules solve only the cases in which the most recent mention is valid. Without lifecycle execution, the learned writer cannot arbitrate versions; without fallback, only capacity-pressure evictions are lost. The complete path is correct in all six scenarios.

0.49. Keeping lifecycle execution but removing fallback loses three capacity-pressure cases, reducing exact match from 1.000 to 0.990. The writer also transfers across disjoint sentence frames (development macro-F1 1.0; shuffled-label control 0.069).

Fallback has a narrow role in this controlled grammar. All three calibrated thresholds reached the minimum of the search grid because the writer was both confident and correct, and keys and values were recoverable by the controlled parser. The low-confidence branch therefore never fired; fallback was used only for the three target slots evicted under capacity pressure, and recovered all three. The multi-hop QA study supplies the complementary setting, with fallback rates of 0.39–0.72.

Generation. Given selected raw lines rather than pre-extracted values, the complete path answers 300/300 with an average prompt length of 146 tokens. The same frozen Llama-3.1-8B [17], when given the entire log, answers 172/300 with 729 tokens. Its errors concentrate on protection and alias resolution, the two operations executed explicitly by the lifecycle. The improvement therefore cannot be explained by shorter prompts alone.

5 Held-Out Multi-Hop QA at Two Lengths

Data hygiene. An early selector had already consumed labels for 20 examples from the local LongBench HotpotQA slice, so that pool is retained only for diagnostics. The confirmatory questions instead come from the official HotpotQA training split [62]. After removing exact and near-duplicate questions against all 500 local LongBench examples, we froze 1,200 training, 300 calibration, and 300 held-out records. Their median context length is 933 words, far below the roughly 7.9k words used by LongBench [3]. We therefore created a second held-out condition with a median of 8,221 words by adding answer-excluded

Reader model	Method	Natural (25% tier)		Extended 8.2k words (10% tier)	
		F1	tokens	F1	tokens
Llama-3.1-8B	Full context	0.714	1477	0.666	12024
	Dense (budget-matched)	0.516	432	0.622	1300
	BM25 (budget-matched)	0.601	437	0.636	1293
	Reader only (no state bound)	0.659–0.663	434	0.695–0.706	1310
	Resident cache + archive fallback	0.659–0.663	434	0.677–0.684	1323
	Oracle evidence (ceiling)	0.717	438	0.786	1301
Qwen2.5-14B	Full context	0.805	1521	0.715	12646
	Dense (budget-matched)	0.594	425	0.753	1338
	BM25 (budget-matched)	0.701	430	0.771	1330
	Reader only (no state bound)	0.758–0.760	428	0.827–0.828	1351
	Resident cache + archive fallback	0.755–0.760	428	0.813–0.830	1367
	Oracle evidence (ceiling)	0.804	432	0.854	1340

Table 5: Held-out multi-hop QA results, with 300 questions in each length condition. Ranges show three fixed selector seeds; token counts are whole-prompt averages. The selector — frozen-Llama hidden-state features, a learned two-hop reranker, a query-independent writer, and calibrated thresholds — was developed with Llama and then applied to Qwen without re-tuning.

distractor passages drawn from the training-record pool. The questions and gold passages remain held out, but the distractor text is train-derived; this is a test of held-out questions among familiar distractor text, not of wholly unseen documents. Before developing the selector, we fixed evidence budgets at 25% of context words for natural records and 10% for extended records. Frozen Llama-3.1-8B and Qwen2.5-14B models serve as readers [17, 47].

For selected passages E and full context x , the intended evidence budget is

$$b(E; x) = \frac{\sum_{p \in E} |p|_{\text{word}}}{|x|_{\text{word}}} \leq \beta, \quad (34)$$

$$\beta = 0.25 \text{ (natural), } 0.10 \text{ (extended).}$$

The first Llama execution used whole-prompt tokens rather than Eq. 34 and is retained as failed under that frozen definition; the corrected Qwen follow-up enforces this equation record by record. Section 8 gives the full accounting.

The comparison with budget-matched dense retrieval was fixed in advance. Across all selector seeds, the learned path improves F1 by 14.3–16.6 points at natural length and 5.5–7.6 points at extended length. The lower end of every paired-bootstrap 95% confidence interval remains positive: at least +8.9 and +1.45 for Llama, and +10.6 and +2.0 for Qwen. That dense feature is a weak frozen-hidden baseline (train-only recall 0.14), so the more informative margin is against BM25: 4.0–6.2 F1. A trained modern retriever, answer-masked and no-evidence controls, and end-to-end archive cost remain necessary before making a general retrieval claim.

The long condition changes the usual cost–quality trade-off. Reading all 8.2k words lowers F1 for both readers (0.714 → 0.666 for Llama and 0.805 → 0.715 for Qwen), consistent with previously reported lost-in-the-middle behavior [37]. Selecting at most 10% of the evidence words instead reaches 102–103% of full-context F1 for Llama and 114–116% for Qwen. No part of the selector was re-tuned for Qwen, yet its margins are larger.

The reader-only rows clarify where the gain comes from. Removing the 32-slot bound does not reduce quality; on the extended set it is slightly better than the cache-plus-fallback path for Llama and roughly tied for Qwen. The learned selector creates the quality margin, while the resident cache preserves most of it. Calibrated fallback fires on 39% of natural and 72% of extended records and searches the retained full context, so total memory is not bounded by 32 slots. Because these passages never overwrite one another, this study supports selection and abstention with bounded hot state, not bounded total memory or the overwrite and protection claims established by the lifecycle study.

6 Why Fallback Is Necessary When Relevance Is Unknown

Figure 11 measures the asymmetry directly. A query-independent salience probe over passage-only features reaches AUC 0.664 on natural records and 0.634 on extended records, only modestly above chance. The query-aware reranker reaches 0.889 and 0.973, although part of the extended-set signal comes from the construction procedure. Under capacity pressure, the 32-slot cache consequently contains only 0.564 of the gold evidence. Without fallback, every missing gold passage is unrecoverable; with a threshold calibrated at the deployment length, most of the selector’s quality is retained. A bounded memory can enforce correct lifecycle behavior for resident information. It cannot promise to store an ordinary fact whose future relevance was unavailable when the write decision was made.

Held-out QA: answer quality at a fraction of full-context active tokens

held-out questions · average whole-prompt tokens

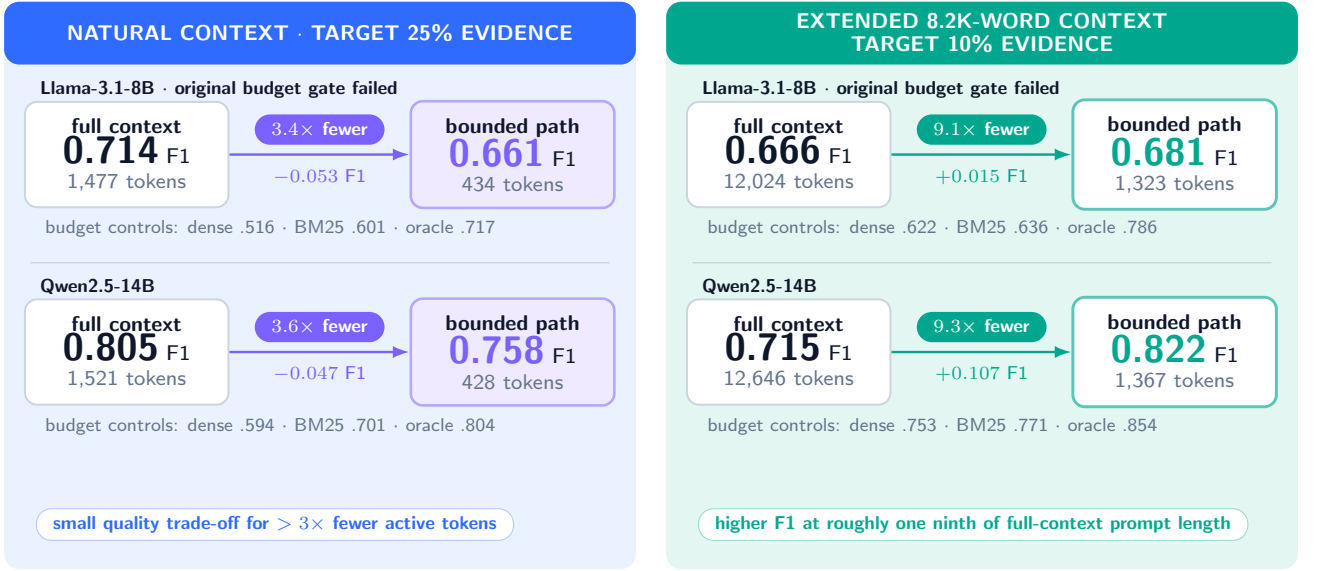


Figure 10: Answer quality and active-prompt cost on the two held-out question sets. Each row compares the resident-cache plus archive-fallback path with the same reader given the full context; budget-matched dense, BM25, and oracle selectors are reported alongside. At natural length, selection trades a small amount of F1 for 3.4–3.6× fewer prompt tokens. At 8.2k words, it improves F1 for both readers while using roughly one ninth as many prompt tokens. These counts exclude archive storage, indexing, retrieval latency, and retrieval energy.

Why sparse fallback is load-bearing

gold-evidence discrimination on multi-hop QA calibration sets

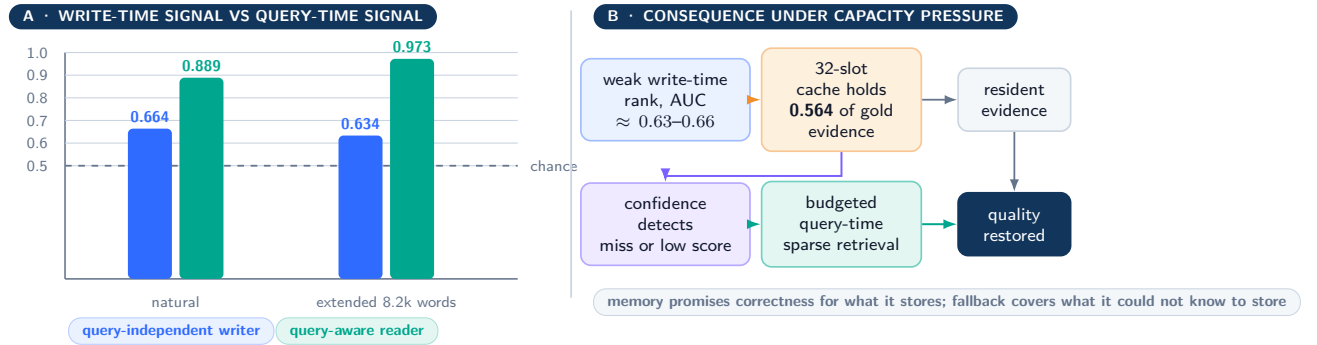


Figure 11: Gold-evidence discrimination on calibration questions. Passage-only features provide little information about which evidence a future question will need. Once the question is available, the same selection problem becomes much easier. A bounded writer therefore needs an explicit way to abstain and retrieve at query time.

7 Relational Binding with Factorized Anchor–Filler Transport

The lifecycle study relies on a deliberately controlled event parser. We therefore ask a narrower trainability question: once event boundaries and semantic role anchors must be predicted, what inductive bias lets a differentiable compiler bind each role to the correct filler under unseen compositions? Pointer networks motivate content-conditioned selection [57], while relational models motivate explicit structural bias [50, 4]. Our first group-disjoint compiler fit every training example but obtained 0/70 held-out successes for role binding and whole-record execution. Optimization was sufficient; the content-matching readout did not compose. This failure motivated AFRT as a structured side path, not as a replacement for all self-attention.

For event e , role r , and candidate filler c , AFRT factorizes the pointer as

$$p(f_r = c \mid x, e) = \sum_a p(a_r = a \mid x, e) T_\theta(c \mid a, r, x, e), \quad (35)$$

where $p(a_r = a \mid x, e)$ is a predicted semantic-anchor posterior and T_θ transports its mass along registered relation paths. Transport depends on relation type, while the two directions and repeated path steps share parameters. Candidate logits are masked by predicted event coverage. They then enter a dynamic partial Sinkhorn assignment P . For the real candidate set C

AFRT: factor semantic anchoring from relation-conditioned filler transport

all four decoders share one encoder, one training trajectory, and one checkpoint

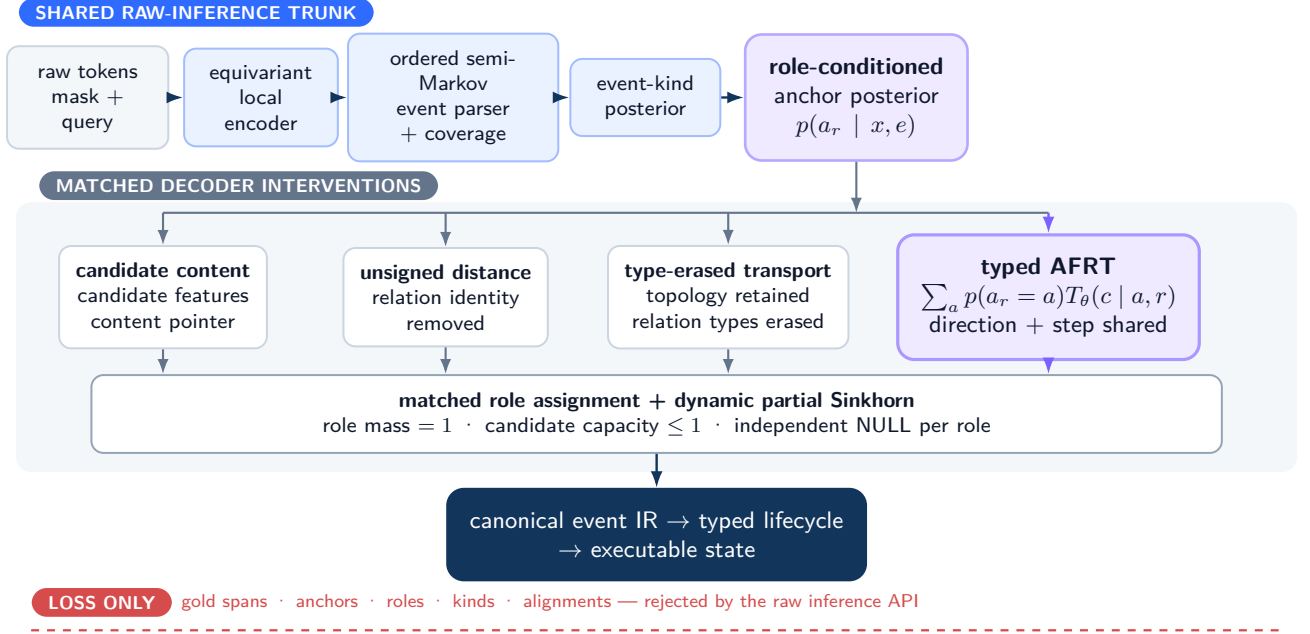


Figure 12: **Relational-binding architecture and matched controls.** The raw encoder, semi-Markov event parser, event-kind head, role-anchor posterior, canonical intermediate representation, and typed lifecycle are shared. The four decoders use candidate content, unsigned distance, type-erased topology, or typed AFRT transport. Direction and path-step parameters are shared in the typed decoder. Gold spans, anchors, roles, kinds, and alignments enter the losses only and are absent from raw inference.

and the private NULL candidate $\emptyset_r$ of each role,

$$\begin{aligned} \sum_{c \in C} P_{rc} + P_{r\emptyset_r} &= 1, \\ \sum_r P_{rc} &\leq 1 \quad (c \in C), \\ P_{r\emptyset_{r'}} &= 0 \quad (r \neq r'). \end{aligned} \tag{36}$$

Thus every role places one unit of mass, real fillers cannot be reused, and each role may independently remain unfilled. The typed decoder receives no filler-content feature, absolute position, signed-direction parameter, template or wrapper identifier, or gold span, anchor, role, kind, or alignment.

Matched controls. Each seed trains one model and publishes one final checkpoint. Four jointly trained decoders are read from it: a candidate-content decoder, an unsigned-distance decoder, a type-erased transport decoder, and the typed AFRT decoder. They share the raw encoder, event partition, coverage, kind and anchor heads, initialization, data order, optimizer, token exposure, and checkpoint. Their paired differences do not mix decoder structure with independently selected checkpoints. They do, however, permit co-adaptation through the jointly trained shared encoder, and none is a type-aware non-factorized control. The present comparison therefore identifies a need for typed topology relative to the registered ablations; it does not isolate factorization as the unique cause.

Richer topology and fixed repetitions. Each of three fresh data seeds contains 512 fitting records and 240 physically disjoint held-out records. The generator varies path length from 2–8, relation composition, distractor bridges, event count from 2–6, and optional or NULL roles. Six stress subsets contain 40 records each. Every model is trained for 32 complete epochs and 512 optimizer steps at effective batch 32. We fixed the three data/model seed pairs before training and did not use early stopping, choose among checkpoints, replace a failed seed, or tune a threshold after seeing a held-out result.

For decoder d and seed s , the paired whole-record gain over a matched control b is

$$\Delta_{d-b}^{(s)} = \sum_{i=1}^{240} (\mathbf{1}[\hat{y}_{i,d} = y_i] - \mathbf{1}[\hat{y}_{i,b} = y_i]). \tag{37}$$

This quantity is positive only when the proposed decoder solves more of the same records than its control.

Result. We counted the hypothesis as supported only if every one of the three fixed seed pairs met every condition in Table 6; no failed seed could be replaced. All three did. Typed AFRT obtains 240/240 whole-record exactness and cross-view

Predefined check	Seed 0	Seed 1	Seed 2	Required result
Each of 11 typed-decoder output checks	480/480	480/480	480/480	$\geq 432/480$
Cross-view consistency	240/240	240/240	240/240	$\geq 216/240$
Whole-record exactness	240/240	240/240	240/240	$\geq 216/240$
Each of six stress subsets	40/40	40/40	40/40	$\geq 34/40$
Gain over candidate-content decoder	+240	+240	+240	$\geq +24$
Gain over unsigned-distance decoder	+240	+240	+240	$\geq +18$
Gain over type-erased transport	+240	+240	+240	$\geq +12$
All conditions met for this seed?	yes	yes	yes	yes

Table 6: Fresh three-seed confirmation on physically disjoint fitting and held-out records. The eleven output checks cover boundary, active anchor, present anchor, NULL anchor, role, kind, canonical intermediate representation, execution trace, state, resident selection, and payload exactness. Every matched control has zero whole-record successes on every seed. These are controlled synthetic results, not natural-language, long-context, or full-language-model results.

AFRT confirmation: one shared checkpoint, four structural decoders

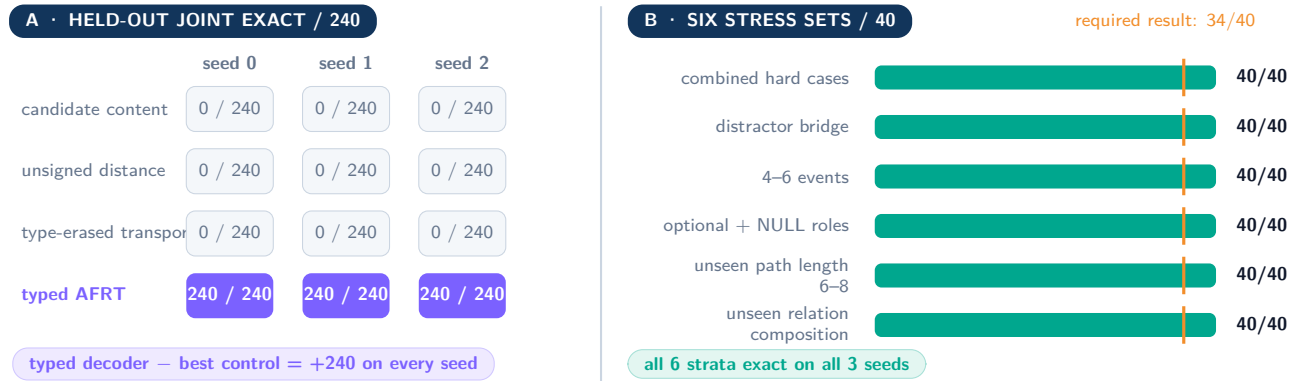


Figure 13: Matched-control result for relational binding. Every decoder is read from the same checkpoint for a given seed. Typed AFRT is exact on all 240 held-out records for each of three fresh seeds; candidate-content, unsigned-distance, and type-erased transport decoders have no whole-record successes. The right panel separates the six predefined stress subsets.

consistency on each seed, and all six stress subsets are 40/40. Each registered control has no whole-record successes, yielding $\Delta = +240$ against every control on every seed. The result supports a narrow causal statement: in this controlled topology and joint-training setup, typed relation-conditioned transport supplies a reproducible binding signal absent from content, unsigned-distance, and type-erased alternatives. A decisive factorization claim still requires a parameter-matched, type-aware non-factorized decoder and independently trained same-initialization/equal-compute repetitions. The result does not establish natural-language relation discovery, arbitrary graph reasoning, long-context scaling, or superiority to external sequence architectures.

Failure and evaluator audit. The evidence did not improve monotonically. The initial group-disjoint compiler failed as described above. A later single-seed qualification isolated an 80/80 typed-transport result against 0/80 for candidate content, but was not enough for a confirmation. The first three-seed richer-topology study remains failed under its original evaluation contract. Two post-hoc audits then found that payload exactness omitted a separately stored lifecycle version and that anchor equality counted inactive padded events. The fresh confirmation used new data and model seeds, with both definitions fixed before training and before any metric was observed.

One evaluation-only recovery is also disclosed. Canonical JSON sorted the metric keys, while the original adjudicator incorrectly required dictionary insertion order. A versioned adjudicator replaced that requirement with exact key-set equality; missing or extra metrics still fail closed. No model, data, prediction row, score, seed, numerical condition, or threshold changed, and the superseded aggregate path remains unused.

Originality boundary. Ordered semi-Markov segmentation [51], Sinkhorn assignment [13], canonical intermediate representations, typed lifecycle execution, pointer-style selection [57], and generic relational inductive bias [4] are inherited tools. The proposed mechanism is the anchor-posterior/filler-pointer factorization in Eq. 35, filler-content-invariant typed transport with direction and step sharing, the partial assignment constraints in Eq. 36, and the same-checkpoint structural intervention. We do not attribute novelty to the standard components that make the mechanism executable.

8 Preregistration History and Forward-Looking Corrections

We report the full preregistration history, including the experiments that did not support the hypothesis.

Lifecycle-task re-registration. The original versioning benchmark accidentally made “take the last line mentioning the queried key” correct on every example. No learned method could exceed that 1.0 baseline by the three points required in the original protocol, and the data never exercised fallback, capacity, or protection. Cross-review found the defect before the test split was opened. We then registered the harder generator in §4; the original test split remains unread.

Evidence-budget definition. The multi-hop QA protocol intended to limit selected evidence to a fraction of the context words. Its frozen implementation instead divided whole-prompt tokens by context tokens, thereby counting roughly 60 fixed question and instruction tokens against every method. Under that implementation, the Llama run exceeded its allowances: 29.4% rather than 26.25% at natural length and 11.0% rather than 10.5% at extended length. We retain that run as failed under its frozen definition even though its quality comparisons were positive on every seed.

The saved prediction rows permit a separate evidence-word reading. The extended set respects the intended 10% limit on every record. Natural contexts use 22.6% on average, but a greedy first-passage exception reaches 61% on one short record. For later work only, we adopted the evidence-word definition and replaced that exception with a hard per-record cap, truncating the top passage only when no whole passage fits. The Qwen follow-up uses this corrected packer and respects both limits exactly. We therefore describe it as a forward-looking follow-up, not as a second successful execution of the original Llama protocol.

Relational-binding evaluator contracts. The first three-seed richer-topology result remains “not demonstrated” under its frozen evaluator. We did not recompute that verdict after finding the payload and padded-anchor defects. Instead, the defects were documented in two post-hoc audits, and a new confirmation used fresh data and model seeds with corrected definitions fixed beforehand. We required all three fixed seed pairs to meet every predefined condition, and they did. The JSON-key-order recovery described in §7 happened before aggregate scoring, wrote to a new path, and changed no scientific quantity. Keeping the failed run, the audits, and the fresh confirmation separate prevents evaluator repair from becoming retrospective model selection.

9 Supporting Controlled Evidence (Corrected)

The integrated path builds on a longer sequence of controlled studies. A logic audit found nine issues, including a fixed target position, two evaluation shortcuts, an entity-identity split overlap, and invalid pooled confidence intervals over repeated use of the same examples. We corrected and reran the affected studies and report repetitions separately.

- **2M-token synthetic stress (collision-free rerun):** explicit memory + sparse fallback scores 49/50 at 2,097,152 tokens with ≤ 132 active chunks; fixed-state and sparse-only proxies each fail complementary scenario classes.
- **Trainability:** a 2.74M-parameter causal event-token backbone learns routing signals (199, 196, and 200 of 200 across three seeds with lite write supervision).
- **Frozen-hidden bridge:** all 18 predefined model–seed comparisons across six model families reach at least 57/60 on entity-disjoint splits, showing that published-model hidden states contain lifecycle signal when key identity is supplied.
- **Controlled grounding and generation:** oracle-free five-digit lifecycle at 90/90 per model/seed (Llama/Qwen); causal writer + hard lifecycle + generation at 90/90; trained soft-token value compression is a documented negative (unstable 85–89/90 across regularizers), with lossless span replay strictly better (90/90/88).
- **Open-domain selector precursor (IDF multi-hop):** a non-learned IDF multi-hop selector recovers structured variable-tracking chains (+58.0 points, CI [49.8, 65.8]) but fails open-domain HotpotQA (−7.1 F1, CI [−14.6, 0.25]), motivating the learned two-hop selector.
- **Differentiable compiler sequence:** the first compiler fits but does not generalize role binding; a single-seed same-checkpoint study isolates an 80/80 typed-transport result against 0/80 for content matching; the first three-seed richer-topology study remains failed; two audits identify evaluator defects; and a fresh three-seed confirmation tests the corrected question (Table 6).

10 Negative Results

All artifacts are retained. Raw frozen-language-model hidden states are weak retrievers (layer-probe recall 0.14; dense-only F1 0.47 at natural length) and become useful only as features in a learned reranker. An IDF-weighted bridge feature and two anchors did not improve that reranker. Presenting evidence in document order rather than score order cost six F1 points at extended length. Calibrating fallback at the wrong context length reduced evidence coverage from 0.88 to 0.31. Trained soft-token value compression never reached the predefined accuracy requirement and was replaced by lossless span replay.

Two earlier results are explicitly superseded or retained as failures. A first scaffold reported 0.02 accuracy without lifecycle control, but its tie-break deliberately preferred stale answers; Table 4 uses a neutral tie-break and obtains 0.49. The first group-disjoint compiler fits its compact training distribution yet has no held-out role-binding successes. The first three-seed richer-topology study also remains failed even though later audits show that part of its apparent payload and anchor failure came from the evaluator. We answer the corrected question with fresh data and seeds rather than rewriting either verdict.

11 Related Work

Efficient sequence backbones. Linear attention, structured state spaces, recurrent hybrids, and long convolutions compress history into a recurrent or convolutional state [29, 12, 20, 54, 45, 19, 15, 46, 61, 24]. Sparse attention limits the set of positions read by each query [64, 63, 10, 16]. Both directions primarily redesign token mixing. We study the semantics imposed on a bounded state when it must support overwrite, protection, eviction, and abstention.

Kernel substrate. FlashKDA is a high-performance CUTLASS implementation of Kimi Delta Attention with token-parallel and head-parallel recurrent updates [11]. Its recurrent state has shape $[B, H, V, K]$; current source documentation reports BF16 or FP32 state, FP32 update accumulation in the optimized path, automatic dispatch through Flash Linear Attention, and H20 kernel speedups over earlier KDA/GDN implementations. Those are upstream kernel measurements, not MMLA measurements. FlashKDA does not define semantic row authority, hindsight supervision, overwrite versus NULL, or lifecycle evaluation. It is nevertheless a plausible complementary substrate for a recurrent/linear MMLA branch if the state geometry and precision contract match. We have not integrated or benchmarked that combination.

Addressable and external memory. Memory Networks and Neural Turing Machines introduced differentiable addressable stores [59, 18]. Recurrent-memory and virtual-context systems carry state across segments or calls [9, 43]. Other Transformer memory systems use recurrence, compression, nearest-neighbor retrieval, retrieved text, or learned memory layers [14, 48, 60, 58, 6, 34, 8, 30, 5, 42]. H2O, Scissorhands, SnapKV, and Landmark Attention retain or index selected KV-cache entries [67, 38, 35, 41]. The closest system-level comparison is sparse memory over static documents (MSA [10]). Our experiments add same-key versioning, protected state, explicit eviction rules, and calibrated abstention when future relevance is unavailable at write time.

Two 2026 systems sharpen the comparison. MemOps is a diagnostic benchmark for remember, forget, update, reflect, and trajectory-level operations in long-horizon conversations [23]. It contributes gold operation traces and six probe families across 403 evidence conversations and 2,006 question–answer pairs; it does not prescribe a memory architecture. Metis is an architecture-level memory foundation model with persistent native state, memory attention, memory-specific mid-training objectives, and gradient-free forward state updates at inference [66]. It is the closest primary-source comparator to the training placement advocated in Eq. 30. Metis also reports declining performance under longer update trajectories and interference from irrelevant active memory, which reinforces the need for explicit overwrite, abstention, and long-running lifecycle evaluation. Our proposal differs by assigning authority to bounded complete rows and by pricing one overwrite against NULL with conditional future risk. We do not claim superiority: MMLA has no learned expected-risk result, while Metis already reports broad memory evaluations.

Work	Problem	Persistent state	Operation	Training	Inference update	Evaluation / unresolved limit
MMLA	Future-use-aware bounded persistence	S authoritative rows plus completed-segment workspace	One complete-row overwrite or NULL per event	Frozen-backbone interface qualification, then grouped-future action values; joint training is conditional	Causal value model selects one row or NULL per event	Controlled component evidence; three interface negatives; no learned predictive-overwrite result
MemOps [23]	Diagnose lifecycle competence	Gold conversational state and operation traces for evaluation	Remember, forget, update, reflect, trajectory composition	No architectural training prescription	System under test executes its own policy	403 conversations, 2,006 QA; benchmark, not a memory mechanism
Metis [66]	Native continually updated model memory	Compressed persistent state accessed by memory attention	Native remember/forget/update/reflect procedures	Memory-specific mid-training with backbone frozen	Gradient-free forward state transformation	Broad memory benchmarks; long trajectories and interference remain difficult
FlashKDA [11]	Fast delta-attention recurrence	Recurrent matrix $[B, H, V, K]$	Numerical recurrent update, not semantic overwrite	Kernel implementation; no memory-policy objective	Fused KDA state transition	Upstream H20 kernel results only; no MMLA integration or lifecycle semantics

Table 7: Primary-source comparison. “Operation” distinguishes numerical state evolution from semantically authoritative memory mutation. The four systems are complementary rather than points on one leaderboard.

Predictive state and consolidation. Predictive state representations describe latent state through predictions of future observations rather than a reconstruction of history [36]. Predictive information formalizes the mutual information shared by past and future [7], and the information bottleneck asks for a compact code that preserves a designated relevant variable [55, 2]. Complementary learning systems separate fast episodic storage from slower integration into distributed parameters [40]. World models and latent-imagination agents learn compressed dynamics for prediction and control [21, 22]. Our construction combines these motivations in a narrower language-model operation: grouped actual futures estimate the conditional action value of one bounded overwrite, then a causal value model predicts those values for deployment. We do not claim novelty for predictive state, value estimation, or consolidation in isolation.

Retrieval and relational binding. The QA study combines BM25 [49] with dense-retrieval-style features [28], evaluates on HotpotQA [62] at LongBench-scale lengths [3], and interprets the full-context comparison in light of lost-in-the-middle behavior [37]. The relational study draws on pointer selection [57], relational modules [50, 4], ordered semi-Markov inference [51], and Sinkhorn normalization [13]. We claim neither the inherited components nor a general graph reasoner; the tested mechanism is their anchor-posterior/typed-transport factorization under same-checkpoint content, distance, and type-erasure controls. RULER supplies complementary long-context stress tests [25]; Llama and Qwen are frozen readers [17, 47], served with vLLM [32].

12 Reproducibility

Every experiment in §4–7 records the runner commit, SHA-256 of every frozen split, generation and optimization seeds, calibration and bootstrap seeds, per-record predictions, and a completion marker. Protocols and later corrections are versioned and were committed before the data read they govern. The controlled versioning seeds are 202607091–93; the ledger-controlled HotpotQA split excludes near duplicates, and the extended contexts use seeds 202607098/99. Selector seeds are 202607102–104 and bootstrap seeds are 20260709/20260710. The fresh relational confirmation uses data seeds 2026080411–13 and model seeds 2026080421–23. Its protocol, freeze receipt, and final aggregate have SHA-256 prefixes c759ac80...462be, 0ca2b847...4688, and e74d92ce...fb9ea. Superseded artifacts remain preserved rather than overwritten.

These hashes and paths provide an internal audit trail, not external reproducibility by themselves. A public release must map every cited digest to an accessible repository revision, data manifest, checkpoint or deterministic constructor, executable environment, and evaluator command. Until that package is published, readers outside the project cannot independently resolve the internal artifact paths in this report.

13 Discussion: Memory Beyond Context Length

Ultra-long context remains a stringent stress test, but it is not the final objective. A capable persistent system should improve its state as evidence arrives rather than keep replaying an ever-growing transcript. The distinction matters for long-running assistants, agents that revisit a changing world, and any model expected to learn during use without immediately changing all of its weights. In this view, memory quality is measured by revision, preservation, deletion, reuse, and future consequence, not by the largest accepted token count.

The architecture also suggests a testable scaling hypothesis. Let $G^*(c)$ be the future-risk reduction available to an oracle memory policy at model capability c , and let $\epsilon_Q(c) = \sup_{s,a} |\hat{Q}_c(a | s) - \bar{J}_c(a | s)|$ be the action-value error. The greedy decision bound in Appendix J gives

$$G_{\text{value}}(c) \geq G^*(c) - 2\epsilon_Q(c). \quad (38)$$

If stronger models both extract more value from correct memories and estimate conditional action values more accurately, $G^*(c)$ may rise while $\epsilon_Q(c)$ falls. Memory could then become more useful as the model becomes more capable, rather than merely compensating for a weak backbone. This is an empirical scaling hypothesis. It is not an AGI theorem.

Persistent, revisable state may nevertheless be an important ingredient of more general intelligence. It offers a middle time scale between transient KV state and slow weight updates: experience can change behavior immediately, remain bounded, be corrected, and later supervise slower learning. That substrate is insufficient on its own. General intelligence also requires goal formation, active exploration, planning, causal world models, social reasoning, calibration, safety, and stable parameter-level continual learning. The appropriate claim is therefore conditional: predictive memory consolidation is a plausible architectural line for persistent adaptive intelligence, and its value can be falsified by the matched and scaling experiments above.

14 Limitations and Next Steps

MMLA roadmap: localize first, open each gate once

all boxes below the registered negative are FUTURE / UNVERIFIED; arrows are stop rules, not promises

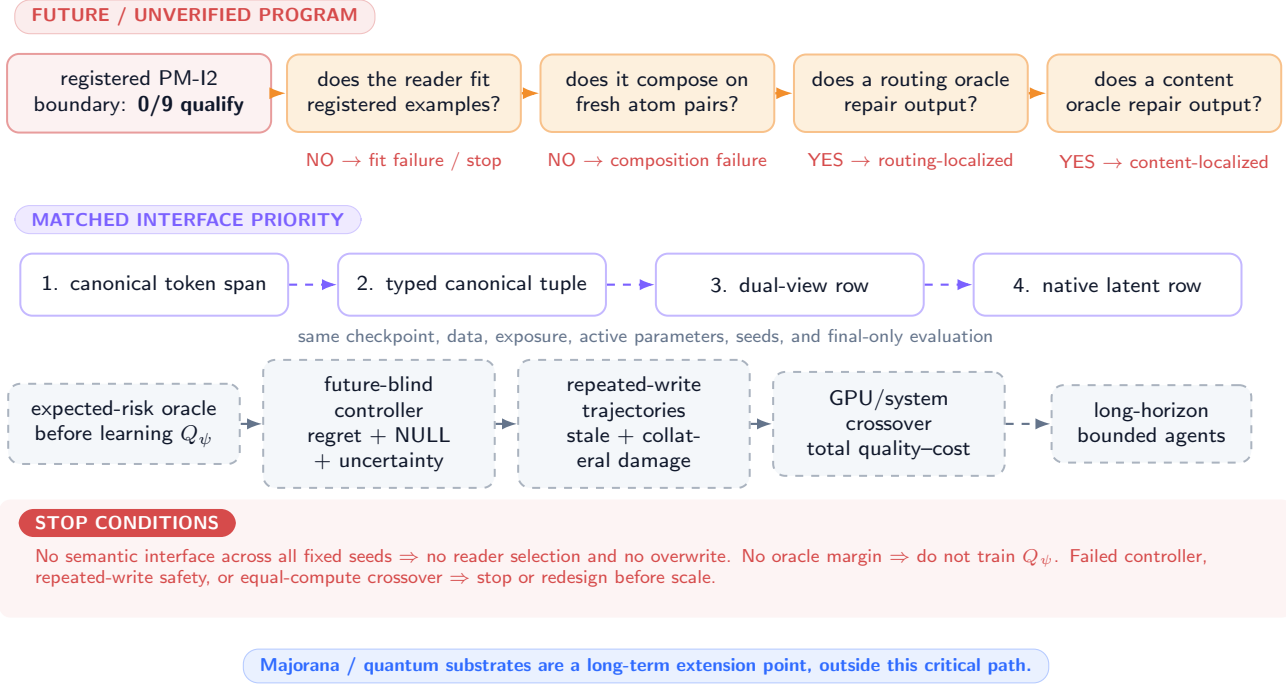


Figure 14: **Conditional roadmap, not an achieved system.** Every dashed successor is proposed or future work. Progress stops when the preceding frozen matched gate fails; Majorana and quantum backends remain outside the critical path.

The lifecycle and multi-hop QA experiments use fixed language models; only the writer, reranker, and thresholds are trained. The versioning language is controlled and shares cue words across otherwise disjoint sentence frames, so it does not establish open-language event parsing. The QA passages are static and exercise only the bounded cache and fallback, not same-key overwrite. Its natural-length budget may also require truncating a passage in very short contexts, and passage-level selection leaves sentence-level budgeting unexplored.

The relational experiment closes one binding failure in a compact synthetic compiler. Its perfect result is tied to the registered generator, seen relation primitives, short sequences, and a jointly trained shared encoder. It is not evidence that AFRT improves a full language model, that factorization is uniquely responsible, or that the model discovers arbitrary relations in natural text. The three negative interface studies in Tables 2 and 3 add a more immediate limitation: physical memory rows can be stable while their semantic content remains unusable to registered lightweight, structured, and native readers. PM-I2 broadens the failure beyond a single external codec, but does not distinguish whether the bottleneck lies in optimization, the stored representation, the compositional output geometry, or their coupling. It does not reject the V28 memory mechanism or MMLA as a whole. The predictive architecture adds further open questions: whether conditional future utility is predictable from the past, whether a local bidirectional encoder helps beyond equal-compute causal encoding, whether repeated atomic overwrites remain stable, whether learned payload construction can be separated from action selection, and whether branch-generation cost is amortized by deployment savings.

Persistent writable memory also creates a security surface that the present experiments do not test. Relevant failures include poisoning a candidate so that it survives consolidation, exhausting slots through malicious writes or locks, cross-user leakage through shared resident state or archives, alias or routing collisions, unauthorized deletion, and rollback that resurrects stale values. A deployable system needs provenance, tenant isolation, write-rate limits, protected-capacity policy, auditable tombstones, deletion verification, and recovery semantics. These are system requirements, not consequences of the information-bottleneck objective.

The immediate next step is not a larger model and not predictive overwrite training. PM-I3 is a fault-localization tree, not a reader-selection contest. Reusing immutable PM-I2 finals without optimizer updates, it asks in order whether failure already occurs on fitted examples, appears only on fresh composition, is repaired by a routing oracle, is repaired by a content oracle, or reaches full qualification. A fit failure stops generalization claims. A fresh-composition failure stops reader selection. Oracle repair localizes a defect but does not qualify the reader. At this revision PM-I3 has no accepted learned output; it is a plan, not a result. Retraining the same nine jobs, changing thresholds, or retrospectively preferring one reader is forbidden.

If that diagnosis justifies a fresh matched interface study, the priority order is canonical token span, typed canonical tuple, the proposed dual-view row, and pure native latent row. Checkpoint, data, exposure, active parameters, seeds, and final-only evaluation must match. Reconstruction, fresh composition, query, calibration, preservation, stale suppression, and mapping must all pass across fixed seeds. Failure of any semantic gate keeps overwrite closed.

Only then may an expected-risk oracle replay all registered slot actions and NULL from the same snapshot. No preregistered oracle margin means Q_ψ is not trained. A future-blind controller must report action regret, top-1 agreement, NULL precision/recall, calibrated uncertainty, stale leakage, collateral damage, and write rate. Repeated-write trajectories test churn, rollback, protected rows, adversarial writes, and long-horizon preservation before any system-scale claim. GPU/system comparison comes last and must include archive storage, indexing, retrieval, branch replay, kernel count, memory traffic, latency, and energy. Majorana and quantum backends are long-term extension points outside this critical path.

15 Conclusion

Memory management asks a different question from context extension: not how much history can be presented, but which parts should be allowed to change persistent state. We formalized one answer. A causal model predicts normally; a completed local segment may be reinterpreted and eventized; grouped true future outcomes estimate the conditional risk of each feasible event-level action; a causal value model chooses one bounded overwrite or NULL per event. The construction is compatible with finite information capacity. That compatibility does not guarantee learnability, efficiency, safety, or intelligence.

The completed evidence establishes useful components. Explicit lifecycle execution resolves overwrite, protection, and eviction in controlled language; calibrated archive fallback handles evidence whose relevance was unknowable at write time; typed transport repairs one controlled relational-binding failure relative to its registered ablations. It also establishes a reproducible bottleneck: three current-state interface studies fail their semantic conditions despite stable physical mapping, and the latest matched dense-row, structured-span, and native-reader comparison remains near chance across all nine fixed jobs. The predictive closed loop therefore remains two evidence gates away: first a semantically usable same-checkpoint reader, then a grouped-future overwrite policy that improves future behavior without increasing interference, compute, or hidden leakage. A positive answer to both would justify joint pretraining and scale studies. The present negative answer rejects the registered interfaces and redirects the next study toward diagnosis before more compute is spent; it does not by itself reject bounded predictive memory.

The architecture is formalized, but its same-checkpoint semantic memory interface and learned predictive overwrite policy are not yet validated.

AI-Assisted Tooling Disclosure

The authors used AI-assisted programming and writing tools for code development, experiment execution, drafting, and editing. The authors reviewed and verified the experimental outputs, claims, and manuscript content and take responsibility for the work.

Author Contributions

Junyi Zou and Avrova Donz contributed equally to this work. Both authors share responsibility for the research and manuscript.

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A Experiment Roadmap

Experiment	Core question
Five-task slot-memory prototype	Can explicit slots solve five synthetic memory tasks?
Token-to-chunk neural interface	Do neural key/value/write signals survive tokenization and chunking?
128K matched architecture comparison	How does the hybrid compare to Delta/GDN/KDA, DSA/NSA, and MSA proxies?
Trainable causal event encoder	Can a causal GRU encoder replace hand-crafted event vectors?
Two-million-token collision-free stress	Does the hybrid hold at two million tokens without key collisions?
Generated-language boundary check	Does generated natural language preserve the synthetic boundaries?
Local long-context benchmark replication	Do Needle/RULER-style local probes replicate the hybrid pattern?
Official benchmark pipeline check	Can the pipeline ingest LongBench/RULER and produce valid predictions?
Frozen-model lexical diagnostics	What do real Llama/Qwen models reveal about simple lexical selectors?
Small trainable memory backbone	Can a 2.74M-parameter model jointly train fast state, slots, and sparse fallback?
Six-family frozen-representation study	Can frozen hidden states from six model families support a memory lifecycle when key identity is supplied?
Learned grounding without oracle keys	Can learned parsers replace oracle key IDs under controlled syntax?
Controlled generation from selected memory	Can selected memory drive frozen-LLM generation, causal lifecycle, soft injection, and learned value-span selection?
First open-domain selector	Does a non-learned IDF multi-hop selector transfer from structured chains to real HotpotQA text? (No: -7.1 F1.)
Controlled lifecycle path	Do query-independent writing, lifecycle execution, and fallback each contribute when lexical retrieval is wrong by construction? (§4)
Held-out multi-hop QA	Does bounded selection beat budget-matched retrieval at two context lengths and with two frozen reader models? (§5)
Relational compiler	Can typed anchor-to-filler transport repair content-binding failure across fresh seeds, richer paths, distractors, multiple events, and NULL roles when all decoders share a checkpoint? (§7)
Predictive-consolidation R1 qualification	Can local consolidation, a detached one-future target, a future-blind controller, and one payload-row overwrite execute against the same completed language-model checkpoint? (Yes mechanically; insufficient for conditional-risk or full-state claims.)
External semantic adapter (PM-S1)	Can a compact external current-state codec recover, query, and preserve the frozen V28 memory state? (No under its registered three-seed contract; mapping invariance and old-value suppression pass.)
Raw representation probe (PM-I1)	Is the frozen hidden representation directly decodable before choosing a denser or more native interface? (No under its registered three-seed contract; downstream interface arms remain unopened.)
Direct interface comparison (PM-I2)	Which of dense row, structured span, or true checkpoint-native readout yields a usable current-state interface under one matched frozen release? (Complete: all nine fixed jobs failed semantic qualification while structural mapping passed; a separately implemented within-project adjudicator published the no-interface terminal.)
Fit-versus-generalization diagnostic (PM-I3)	Does the PM-I2 failure already appear on fitted examples, emerge only on fresh composition, or disappear under routing or content oracles? (Planned against immutable PM-I2 finals; no learned result and no accepted production execution at this revision.)
Same-checkpoint expected-risk intervention	After an interface qualifies, do identical-prefix grouped futures and action-conditioned replay teach a deployable value model that reduces future risk and overwrite interference relative to equal-compute immediate and local/current-risk writers? (Closed because no PM-I2 interface qualified.)
Matched from-scratch pretraining	If the intervention succeeds, does predictive consolidation improve quality and cost across model scales and long-running natural streams at equal token and hardware budgets?

Table 8: Experiment roadmap in execution order. Mechanical qualification, semantic-interface identification, conditional-risk overwrite learning, and from-scratch scale studies are deliberately separated. Scale remains conditional on both a usable interface and a deployable gain in the corrected intervention.

B Extended Related Work

Linear attention and efficient sequence models reduce attention cost by replacing quadratic softmax attention with recurrent, kernelized, state-space, or long-convolution states [29, 12, 20, 54, 45, 19, 15, 46]. Delta-style models and recent hybrid architectures improve the expressiveness and update dynamics of fixed states [61, 24, 31]. VLA frames linear attention as stable associative memory [44]. Our focus differs: we do not only stabilize the state; we add explicit memory lifecycle management outside the fast state.

Long-term and neural memory systems augment sequence models with differentiable stores, recurrence, memory layers, or retrieved memories [59, 18, 14, 48, 9, 60, 58, 6, 5, 42, 43]. RAG-style and nearest-neighbor methods retrieve external text or hidden states [34, 8, 30]; KV-cache methods retain high-value tokens or landmarks during inference [67, 38, 35, 41]. These works motivate memory augmentation; our studied unit is request-local editable memory with explicit overwrite, protection, eviction metadata, and calibrated abstention executed in the evaluation path.

Sparse attention systems reduce the attended set through fixed or learned patterns, from BigBird’s structured sparsity to NSA’s hierarchical selection, DeepSeek’s learned sparse indexer, and MSA’s document/chunk memory retrieval [64, 63, 16, 10]. RACE replaces softmax similarity with random projections and soft locality-sensitive hashing [27]. Sparse retrieval complements explicit lifecycle management: it is useful when relevance is unavailable at write time, but does not by itself define versioning semantics. RULER and LongBench pressure-test these distinctions [25, 3]; HotpotQA supplies multi-hop supervision [62].

C Controlled Lifecycle Protocol Details

Scenarios. Each record has a target key with 2–4 versioned updates, distractor-key updates, and filler prose; sentence frames are disjoint across train/dev/test while cue phrases (update, mention, reject, lock, and alias markers) are shared — the disclosed controlled-language boundary. Scenario semantics: *alias-query* defines the alias once (“*X is an alias for Y*”) and the question names only the alias, which never appears in update lines; *stale-remention* appends audit lines (“*an earlier revision listed Y as v*”) after the final update; *invalid-write* appends rejected change requests with fresh values; *protected-slot* freezes the key at version k (“*locked at v_k ; subsequent writes are void*”) followed by later void updates; *slot-pressure* writes 48 distinct keys against the 32-slot bound. Generator unit tests assert that the lexical last-key baseline returns exactly the designed wrong value (the stale, rejected, or void value, or nothing for aliases) on the four breaking scenarios.

Writer. Multinomial logistic regression over hashed bag-of-words features of the line with keys, values, and digit runs masked (256 dimensions; class-weighted SGD). The feature function accepts only the line text; a unit test asserts two lines differing only in keys/values/step numbers produce identical features. Dev macro-F1 is 1.0 across split-disjoint frames; a shuffled-label control collapses to 0.069.

What counted as support. Before generating the data, we required the complete path to beat the strongest budget-matched hand-written rule by at least three exact-match points on every seed, with the lower end of a paired-bootstrap 95% confidence interval above zero. Its stale, rejected, and void-answer rate also had to be at most half that of the learned writer without lifecycle execution. Accuracy on protected values had to reach 0.90. Fallback had to activate on at least 80% of memory misses and recover at least 90% of the capacity-pressure accuracy achieved by the unbounded lexical rule. Finally, occupancy could never exceed 32 and the writer interface had to remain query-independent.

Before the single held-out read, training and calibration checks also required adequate headroom, failure under shuffled labels, parser precision and recall of at least 0.99, successful one-batch overfitting, no split overlap, and evidence that every component was actually exercised. The runner was tested end to end on calibration data, committed, and then executed once. A completion marker makes a second execution refuse.

D Held-Out Multi-Hop QA Protocol Details

Contamination control. The history ledger records every LongBench record ever touched by earlier phases and freezes the local LongBench HotpotQA pool as diagnostic-only (20 early label-tuned record IDs are unrecoverable). Holdouts derive from the official HotpotQA training parquet: 90,447 rows scanned; 13 in-split duplicates, 0 exact and 2 near-duplicate (token-Jaccard ≥ 0.8) question overlaps against all 500 local LongBench questions excluded; 1,200/300/300 train/dev/holdout frozen by seeded hash rank.

Length-matched extension. Each holdout/dev record is extended to $\geq 8,000$ words by inserting whole distractor passages drawn from *other* train records (pool of 11,713 deduplicated passages), excluding any passage whose normalized text contains the record’s normalized answer (token-boundary match) or whose title collides with an original passage; original passages, including gold, are shuffled to seeded random positions. Original passages and supporting sentences are verified verbatim-present per record.

Selector stack. Dense features are frozen Llama-3.1-8B layer-32 hidden states (attention-masked mean pooling, L2-normalized; the layer was chosen by a train-only retrieval probe — raw hidden states are weak retrievers, recall 0.14, and act only as features). The reader is a logistic reranker over (dense cosine, BM25, lexical overlap, title-in-query, position, length) with a second hop round anchored on the top-1 pick (max-cosine and title-link bridge features), trained on train gold-passage labels, three seeds. The query-independent writer scores passages from position/length/capitalization/digit/title statistics only and fills 32 slots.

Calibration in the deployment length regime. The fallback threshold (confidence = mean of top-2 reader scores over memory) is chosen on a 200-record *train-derived extended* slice: calibrating at natural length, where 32 slots hold all ~ 10 passages, yields a threshold at which fallback never fires and extended coverage collapses from 0.88 to 0.31 — a recorded negative. The slice’s natural-length labels were seen by the reader during training; the direction of this disclosed impurity is conservative (inflated calibration confidence produces *more* fallback at deployment, preserving quality at extra retrieval cost).

Evidence budgets and per-seed results. The budgets were fixed from calibration measurements logged before the selector existed. At 25% of an 8.2k-word context, all selectors nearly saturate evidence recall (BM25 0.970); at 10%, BM25 reaches 0.867 and the learned two-hop selector 0.956. A 10% budget is too small at natural length, where even oracle selection falls to F1 0.60, so that condition uses 25%.

For Llama, the three seed-wise gains over dense retrieval are +14.64/+14.44/+14.31 F1 at natural length, with confidence-interval lower ends +9.38/+8.99/+8.94, and +5.98/+6.27/+5.52 at extended length, with lower ends +1.66/+1.96/+1.45. Qwen gains +16.55/+16.48/+16.04 and +7.10/+7.64/+5.99, with lower ends at least +10.6 and +2.00. Relative to reading the entire context, Llama retains 92.8/92.5/92.4% of F1 at natural length and reaches 102.3/102.8/101.7% at extended length. Qwen reaches 94.3/94.3/93.7% and 115.2/116.0/113.7%.

The two budget readings. Under the frozen whole-prompt-token definition, the Llama run uses 29.4% against a 26.25% allowance at natural length and 11.0% against 10.5% at extended length, so it fails that requirement. Counting selected evidence words instead, the extended per-record maximum is exactly 10.00%. Natural records average 22.57%, but the original packer’s first-passage exception reaches 61.36% on one record. The forward-looking Qwen follow-up uses a hard per-record cap and truncates the top passage only if no whole passage fits; both length conditions then remain at or below their stated evidence budgets.

E Relational-Binding Protocol Details

Data geometry. The fresh confirmation uses data seeds 2026080411–13 and model seeds 2026080421–23, all disjoint from the failed earlier three-seed study. Each seed contains 512 *train_fit* and 240 *fresh_train_heldout* records with unique record and group namespaces across fitting data, held-out data, and seeds. The held-out records are balanced into six 40-record subsets: combined hard cases, distractor bridges, 4–6 events, optional or NULL roles, unseen path lengths 6–8, and unseen relation compositions. Relation primitives are shared across fitting and held-out data. The question is whether seen primitives compose in new ways, not whether the model discovers unseen

relation semantics.

Shared model and four decoders. A translation-equivariant local encoder feeds an exact-count ordered semi-Markov event parser [51], predicted event coverage, an event-kind head, and a role-conditioned anchor posterior. Candidate-content, unsigned-distance, type-erased transport, and typed AFRT decoders are trained jointly and read from one model-only checkpoint per seed. Dynamic partial Sinkhorn assignment inherits differentiable matrix scaling [13]; we do not claim Sinkhorn itself as novel. Raw inference accepts tokens, masks, and queries only. Gold spans, anchors, roles, kinds, and cross-view alignments are loss-only inputs and are rejected by the raw API.

Training and what counted as support. Training consists of 32 complete fitting epochs, 512 AdamW steps [39], and effective batch 32, without a scheduler, dropout, early stopping, or checkpoint choice. For each seed, every one of eleven typed-decoder output checks had to reach 432/480; cross-view consistency and whole-record exactness each had to reach 216/240; and every 40-record stress subset had to contain at least 34 exact records. The paired whole-record gains over the candidate-content, unsigned-distance, and type-erased decoders had to reach +24, +18, and +12, respectively. We would regard the overall hypothesis as supported only if all three fixed seeds met every condition, with no seed replacement.

Evaluator definitions and artifacts. Payload exactness compares both the resident payload and the separately stored lifecycle version. Anchor exactness is restricted to active events and reports present and NULL roles separately; inactive padding is excluded. For repository lookup, the fresh confirmation is labeled R5-P5, and its four decoder identifiers are C0, D0, U0, and R1. The protocol and freeze-receipt SHA-256 prefixes are c759ac80...462be and 0ca2b847...4688. The evaluation-only key-order recovery has protocol and receipt prefixes 37889c87...fa23e5 and 1a9c6756...c5df0. The final aggregate is experiments/phase_a_single_4090/r5_p5_clean_confirmatory_v1/paired_summary_r1.json, SHA-256 e74d92ce...fb9ea. It is complete, all three fixed seeds meet every condition, and both Dev/Test touched flags are false. The evidence-package manifest contains the full digests.

F Audit Summary

An internal logic audit preceded v2. The earliest associative-recall task always placed the requested fact first and assigned uniform importance, making its headline result trivial. Randomizing the target and supervising writes produced real evictions that scale with difficulty. Another evaluator silently gave full context to a supposedly bounded method, while a separate baseline matched generator templates with hand-tuned regular expressions; both are now named as non-memory controls. We also removed entity overlap from the frozen-hidden split, replaced pooled Wilson intervals over repeated use of one fixed set with per-seed reporting, and eliminated filler/target key collisions from the two-million-token generator. The corrected rerun obtains 49/50.

The relational study has its own preserved audit chain. The first three-seed payload definition omitted lifecycle version, whole-view anchor equality counted inactive padding, and the fresh confirmation's first aggregate attempt required JSON dictionary insertion order. The original three-seed verdict remains failed, the first two findings are explicitly post-hoc, the fresh confirmation uses new seeds and pre-corrected definitions, and the key-order change is versioned and evaluation-only. Superseded artifacts are marked and retained. The publicly visible v1 predates these corrections; this version updates its numbers without rewriting historical verdicts.

G Reproducibility Notes

Every run artifact records its git commit; frozen splits carry SHA-256 manifests; one-shot evaluations write completion markers that make their runners refuse re-execution. Mainline commands:

```
# Controlled lifecycle: data, calibration, one-shot holdout, generation
python -m phase14e_open_domain_grounding.phase14e_b_versioned_vt2
python -m phase14e_open_domain_grounding.phase14e_b_track_a_dev2
python -m phase14e_open_domain_grounding.phase14e_b_track_a_test2 \
--split-label test
python -m phase14e_open_domain_grounding.phase14e_b_track_a_generation_dev2 \
--dev data/phase14e_b/phase14e_b_versioned_vt2_test.jsonl --allow-test

# Held-out multi-hop QA: data, embeddings, selectors, Llama, Qwen
python -m phase14e_open_domain_grounding.phase14e_b_manifest
python -m phase14e_open_domain_grounding.phase14e_b_hotpot_extended
python -m phase14e_open_domain_grounding.phase14e_b_embed
python -m phase14e_open_domain_grounding.phase14e_b_track_b_dev2
python -m phase14e_open_domain_grounding.phase14e_b_track_b_dev3 --generate
python -m phase14e_open_domain_grounding.phase14e_b_track_b_confirmatory
python -m phase14e_open_domain_grounding.phase14e_b_track_b_confirmatory \
--model-id qwen2.5_14b_instruct --strict-packer \
--gb3-mode evidence_words --bootstrap-seed 20260710
```

Principal artifacts (all with per-record prediction rows):

- phase14e_open_domain_grounding/results/phase14e_b_track_a_test2_v1/
- phase14e_open_domain_grounding/results/phase14e_b_track_a_generation_test2_v1/
- phase14e_open_domain_grounding/results/phase14e_b_track_b_confirmatory_v1/
- phase14e_open_domain_grounding/results/phase14e_b_track_b_confirmatory_qwen_v1/
- phase4_competitive_baselines/results/sparse_hybrid_replicates_collision_free_20260702/
- phase13_frozen_hidden_memory/results/adapters_disjoint_all_six_20260702/
- phase_stats_correction/results/corrected_current/

Preregistration documents appear in commit order: the mainline plan, the amendment that replaced the defective lifecycle task, the multi-hop QA protocol freeze, and the forward-looking evidence-budget correction. Each predates the data read it governs.

For compatibility with repository filenames, the controlled lifecycle and multi-hop QA studies retain the historical labels Track A and Track B. Relational-binding reproducibility is bound by PHASE_A_R5_P5_CLEAN_CONFIRMATORY_PROTOCOL.md, PHASE_A_R5_P5_FREEZE_RECEIPT.json, and the versioned typed-decoder adjudicator protocol and receipt. The evidence package paper_evidence_r5p5_20260803/ contains the result table, terminal audit, claim boundaries, artifact manifest, and checksums. Per-seed commands, data hashes, model-only checkpoints, prediction rows, summaries, and logs remain in those receipts rather than being abbreviated here.

H Interface Qualification and Conditional Expected-Risk Protocol

Checkpoint binding. The current reference implementation uses the final model-only state from the completed V28 real-data pilot, SHA-256 98c1be40...ab82f5. Its sidecar, distributed-checkpoint tree, configuration, and training freeze are deep-hashed before loading. All 295 tensor keys load exactly with no missing or unexpected key, and every backbone parameter remains frozen. The project’s “160M” label describes the hidden-stack configuration; the serialized model contains 352,347,121 parameters including the large vocabulary interfaces. This distinction is recorded to prevent an architecture-size label from being mistaken for an exact parameter count.

Mechanical ancestor and its boundary. A parameter-free memory reader is exactly identical to the frozen language-model head when persistent state is empty. The local consolidator sees only a completed 64-token segment. The ancestor implementation evaluates each registered slot and a private NULL action from one immutable snapshot, detaches a target, rejects future-bearing deployment arguments, and changes one payload row or none. Thirty-five focused CPU tests and a one-record finite-gradient probe qualify software plumbing only. The code uses one realized continuation, converts its risks into a Gibbs policy, and applies memory after a cached future representation. It therefore neither estimates Eq. 19 nor identifies the effect of an action on future hidden-state evolution.

PM-S1 and PM-I1 terminal interface evidence. PM-S1 trains the external semantic adapter at three fixed seeds, then opens current-state qualification once. It publishes 393,216 raw rows. Independent reduction matches the producer aggregate byte for byte (SHA-256 6167308d...994995) and yields Table 2; utility and predictive adjudication remain unopened. PM-I1 trains a raw-representation probe at three new fixed seeds. Its one-shot aggregate (SHA-256 c0ff1fb3...26d5d) records the gate vector (0, 0, 0, 0, 0, 1, 1) for every seed: all semantic gates fail, both mapping gates pass. The registered terminal decisions are retained rather than repaired or rerun.

PM-I2 direct-interface status. PM-I2 compares D1 dense row, S1 structured 4×192 span, and N1 true checkpoint-native readout with active parameter counts 1,541,377/1,496,577/1,460,737. It fixes three seeds, one Latin order, one hidden release, 2,048 updates per job, and final-only checkpoints. Data and materialization manifests have SHA-256 prefixes c214f467...b5a8 and 960c8ada...b09d5; the accepted hidden manifest is 6c0fcc56...5e08c. The original R2 implementation stopped before update one on the conceptual/physical key mismatch and remains rejected. R3 repaired only that reader schema. Its formal manifest/protocol/receipt prefixes are c790f820...f337c, f9cf7869...f475, and bddc3811...e8b05; its training launch receipt/log are 353d2a1c...27037 and cd1f5aea...65d57. All nine fixed jobs then completed exactly 2,048 updates. Truth-free Phase A was frozen under receipt f024fb05...96b4. Phase B was frozen under receipt 3f33147a...f3efc, opened evaluator truth 86d6f1a3...f242f only after 45 factor files were fixed, and ended with log ffb22a84...f48e5. Nine checkpoint bundles, nine factor bundles, nine metric bundles, and 55,296 raw qualification rows were independently audited. Every job is unqualified: candidate exactness is 0–45/23,040, query exactness 10–1,536/9,216, and mean record-macro Brier 0.998676–0.999093. Old-value suppression and mapping invariance pass. The first adjudication transport stopped before production on a lifecycle-test error and remains preserved under receipt 195e8b02...46e3. R01 changed only launch identity and that test lifecycle, then called the frozen adjudicator once under receipt b6e4a1e9...4778. The final bootstrap/aggregate hashes are 1cb09ce3...b6f and 65ba2236...79ee; the former records not-applicable with zero draws, and the latter publishes pm_i2_no_current_interface_qualified_stop. These accepted measurements and terminal close PM-I2 without selecting an interface.

Post-PM-I2 diagnostic boundary. PM-I3 is designed to reuse the nine immutable PM-I2 final checkpoints without training updates and to separate fit performance, fresh qualification performance, routing-oracle behavior, and content-oracle behavior. Those contrasts may localize optimization, routing, representation, or compositional readout failure, but they cannot rescue a PM-I2 qualification decision or select an interface retrospectively. No PM-I3 learned artifact is admitted in this revision, and its production execution remains pending a separate accepted transport boundary.

Conditional expected-risk successor, before any result. This stage is now closed by the PM-I2 qualification rule: no interface qualified at all three seeds. Had an interface qualified, each future experiment group would have contained one identical observed prefix and four to eight continuations with frozen weights. Group identifiers, not individual branches, would have been physically separated across fitting, calibration, and fresh held-out namespaces. The generator would remove direct lexical cues to the correct write and include strict replacement, stale evidence, preservation of unrelated facts, deletion, delayed relevance, multi-hop consequences, ambiguous prefixes, and groups for which NULL is optimal in expectation. From one immutable snapshot, every action would be scored by replaying every continuation through the frozen reader under the corresponding post-action memory. No further write would be permitted within the horizon. These rules remain a preregistration template, not an authorized experiment.

The no-op, immediate/current-risk, local/current-risk, and local/expected-risk arms will share the V28 checkpoint, slot count, registered actions, candidate payload, fixed gate $g = 1$, write budget, train exposure, optimizer family, and evaluation budget. The local/current-risk to local/expected-risk contrast changes the supervision target only. All trainable arms must be parameter matched; all feasible actions receive value targets. A non-deployable empirical expected-risk oracle is reported only as a ceiling. Payload learning, gate learning, recurrent rollout policy, index learning, and multi-write planning are deliberately outside this successor.

Falsification before scale. The direction stops if the empirical expected-risk oracle has negligible margin over current-risk writing, if the future-blind value model cannot recover a useful action ordering across fresh seeds, if lower future loss is bought by more stale leakage or collateral damage, or if gains disappear under parameter/compute matching. No checkpoint selection, failed-seed replacement, future-dependent deployment input, or post-held-out horizon change is allowed. A successful intervention authorizes a short joint-training study; it does not directly authorize a larger model or an AGI claim.

I Frozen-Hidden-Bridge Protocol Details

The six-family frozen-hidden bridge is a controlled representation study, not a learned or open-domain parser. Each generated sample supplies event sentences, a query sentence, integer `key_ids`, integer `value_ids`, a target event index, and optional write labels. The frozen backbone mean-pools last-layer hidden states over each complete event and query sentence. Separately, the protocol forms the canonical string `entity N` from each supplied key identifier, encodes it with the same backbone, and mean-pools that string. Event and query representations concatenate full-sentence vectors with the canonical-key vectors. No runtime regular expression, named-entity recognizer, exact-string span matcher, or value-span extractor is used; same-key replacement and branch arbitration use exact integer equality. The study therefore asks whether frozen hidden representations support a lifecycle when key identity is supplied. It does not establish entity discovery, alias resolution, or coreference, which the learned QA selector addresses only at passage level.

J Mathematical Properties and Stability Boundaries

Causal boundary. At boundary b_n , z_n is a deterministic or stochastic function of $(X_{\leq b_n}, M_{n-1})$. The updated state M_n is used only for positions greater than b_n . Therefore the conditional distribution for any already emitted token is unchanged by consolidation. Bidirectionality within $(b_{n-1}, b_n]$ does not introduce a path from $X_{>b_n}$ to $p(X_t | X_{<t})$ for $t \leq b_n$. The implementation enforces this property structurally rather than relying on a loss penalty.

Atomicity. Let $\Delta_{n,j}^v$ be the difference between matrices formed by stacking the neural payload views of $M_{n,j}$ and $M_{n,j-1}$. From Eq. 13, $\text{rank}(\Delta_{n,j}^v) \leq 1$ and $\|\Delta_{n,j}^v\|_{0,\text{row}} \leq 1$ for every event. Across the full segment, $\|V(M_n) - V(M_{n-1})\|_{0,\text{row}} \leq K_n$. The authoritative-state definition is stronger: every mutable canonical, neural, routing, version, security, tombstone, and rollback field of the selected row commits together. Any single event trace that modifies two resident rows, or silently mutates an external authoritative alias/index state, is outside the contract. Protection can only remove actions from $\mathcal{A}_{n,j}$; it cannot add a hidden second write.

Resident boundedness. Equation 2 is a bit-level contract, not merely a fixed row count. Every resident field has a fixed maximum width and there are exactly S cells, so $\text{bits}(M_n) \leq B_{\text{resident}}$ for every n . Monotone versions use bounded integers with an explicit fail-closed exhaustion policy: the last representable version cannot be incremented, and counters never wrap, saturate, or reuse identity/version pairs. Canonical content, aliases, and metadata that exceed their caps likewise make the candidate infeasible. Unbounded provenance, rollback history, archives, indexes, and audit events are external objects referenced by fixed-width digests and charged to A_n . Thus resident state is truly bounded while total retained system information may grow.

Dual-view consistency. A row is valid only when Eq. 11 holds and Eq. 10 binds one identity and version to both canonical and neural bytes under one frozen encoder version. Hashes are receipts rather than the equality predicate. The receipt is committed atomically with the row, and byte equality plus the receipt are rechecked at load, recovery, export, teacher replay, and index rebuild. A mismatch cannot be repaired by selecting one view: the row is quarantined, removed from the feasible read/write set, and requires a new validated version. This provides mechanical byte consistency and a fail-closed semantic qualification boundary; it does not prove that the learned neural view is semantically adequate.

Multi-event teacher semantics. Equations 15–17 define each event value on its own branch prefix and include later event decisions in the cost-to-go. Exact dynamic programming, a frozen continuation policy, and the registered $K_n = 1$ intervention are distinct protocols and must be labeled as such. The policy minimization occurs outside the future expectation; a per-future argmin would be a clairvoyant teacher unavailable to deployment. Reusing the pre-segment candidates at later events, averaging event-local argmins, or scoring each event while silently holding later writes undefined is not a valid teacher. Grouped future weights are fixed before outcomes are opened, and group identity is the data-splitting unit.

Complete-candidate accounting. Equations 28 and 29 charge semantic proposal, trusted assembly, dual-view validation, action scoring, commit, branch-prefix expansion, and future replay. Any target shortlist or shared constructor must publish its own construction, miss, and excluded-action costs. Consequently, the simpler $O(TSd)$ read expression is not an end-to-end MMLA cost bound.

Entropy-regularized optimum for a fixed expected-risk vector. Expanding Eq. 26 gives

$$\mathcal{F}[q] + \tau \log Z = \tau \sum_a q(a) \log \frac{q(a)}{q^*(a)} \geq 0.$$

Equality holds exactly at $q = q^*$. The Gibbs distribution is therefore the unique regularized minimizer when $E(a) = \bar{J}_H(a | s)$ is fixed, every registered action has finite risk, and $\tau > 0$. The identity neither estimates $\bar{J}_H$ nor permits the expectation to move through softmax. The future expected-risk successor estimates the conditional values first and uses direct value regression.

Greedy action-value regret. Let $a^* = \arg \min_a \bar{J}_H(a | s)$ and $\hat{a} = \arg \min_a \hat{Q}(a | s)$. If $\max_a |\hat{Q}(a | s) - \bar{J}_H(a | s)| \leq \epsilon$, then

$$\bar{J}_H(\hat{a} | s) - \bar{J}_H(a^* | s) \leq 2\epsilon.$$

Indeed, $\bar{J}_H(\hat{a}) \leq \hat{Q}(\hat{a}) + \epsilon \leq \hat{Q}(a^*) + \epsilon \leq \bar{J}_H(a^*) + 2\epsilon$. This bound motivates held-out value error and action regret. It does not guarantee that a finite grouped benchmark estimates the true deployment distribution; that requires representativeness and fresh groups.

Long-horizon boundary. Authoritative rows do not obey a soft global-convergence claim: a newer fact causes a discrete hard commit, and routing, thresholded writes, and eviction form a switched process. A persistent false-write rate can therefore bias or oscillate even though each row and update is bounded. The predictive teacher is intended to reduce this error; only a long-horizon overwrite benchmark can establish repeated-write stability.